

JEE MAIN & ADVANCED

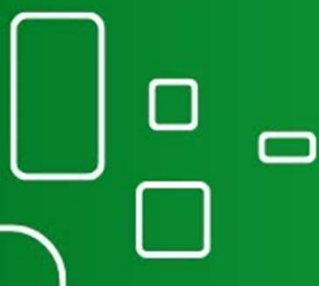
CLASS 12



MATRICES

FULL CHAPTER | ONE SHOT

MATHS | ARVIND SIR



Nature of Chapter:

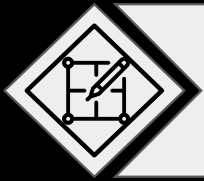
- 1) It is an Easy and Scoring Chapter.**
- 2) It has a lot of Definitions.**
- 3) Major questions, you will find easy to average if you study topic carefully.**
- 4) Rarely you will find above average questions, either it will be upto average level or very tough level.**
- 5) If you want to do good in this chapter, then practice the Properties of all topics a lot.**

Weightage of Determinants (Last 5 years)

	2023	2022	2021	2020	2019	Average
JEE Main	2.2 %	2.3 %	3.6 %	5 %	5 %	3.62 %
Jee Advanced	3 %	3 %	3 %	0 %	3 %	2.4 %

Weightage of Matrices (Last 5 years)

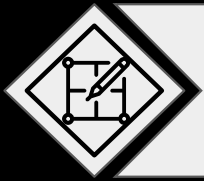
	2023	2022	2021	2020	2019	Average
JEE Main	4.2 %	4.7 %	4.1 %	2.8 %	1.9 %	3.54 %
Jee Advanced	6 %	5 %	7 %	5 %	9 %	6.4 %



MATRICES

- Describing Matrices
- Types of Matrices
- Algebra of Matrices
- Relation between Matrices and Determinants
- Transpose and its properties
- Some special types of Matrices
- Adjoint & Inverse of Matrices
- Elementary Operations
- Solving system of linear equations

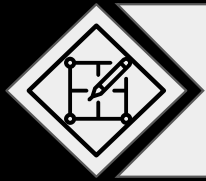
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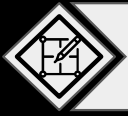
MATRICES

Critical Topics in the Chapter

- Algebra of Matrices
- Transpose and its properties
- Adjoint & Inverse of Matrices



Describing Matrices

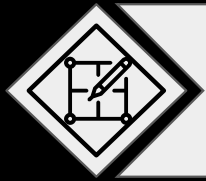


Describing Matrices

Matrix

It is an arrangement of $m \times n$ elements in 'm' rows & 'n' columns.

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \quad m \times n$$



Types of Matrices



Types of Matrices

(a) Row matrix:

$$[2 \ 3 \ 0]_{1 \times 3}$$

(c) zero or Null matrix:

$$a_{ij} = 0 \quad \forall i, j$$

(e) Vertical matrix:

$$\text{No of rows} > \text{No. of cols}$$

(b) Column matrix:

$$\begin{bmatrix} 2 \\ 0 \\ 0 \\ 1 \end{bmatrix}_{4 \times 1}$$

(d) Horizontal matrix:

$$\begin{bmatrix} 2 & 1 & 3 \\ 0 & -1 & 2 \end{bmatrix}_{2 \times 3}$$

$$\text{No of cols} > \text{No of rows}$$

(f) Square matrix:

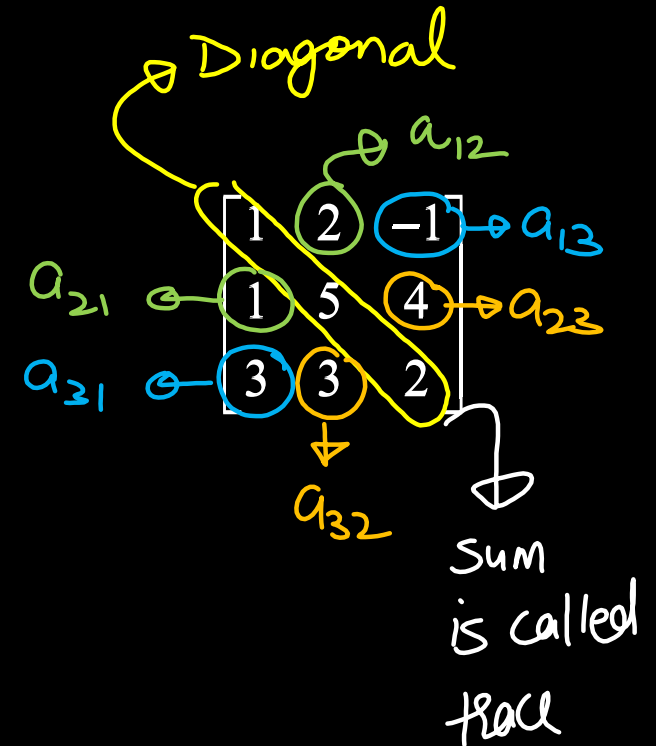
$$m = n$$



Types of Matrices

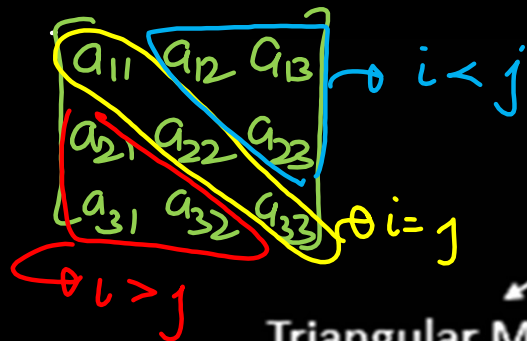
In square matrix:

- (i) a_{ii} are called diagonal elements.
- (ii) a_{ij} & a_{ji} are called conjugate elements.
- (iii) $\sum_{i=1}^n a_{ii}$ is called trace of square matrix.





Types of Matrices



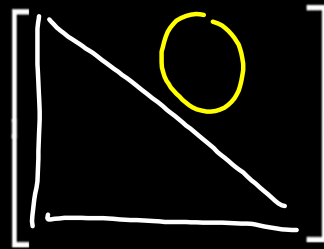
Triangular Matrix

Square Matrix

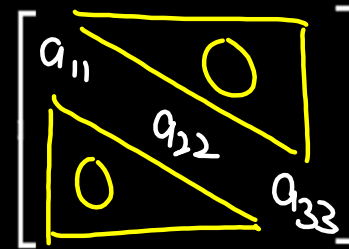
Diagonal Matrix

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ 0 & a_{22} & a_{23} \\ 0 & 0 & a_{33} \end{bmatrix}$$

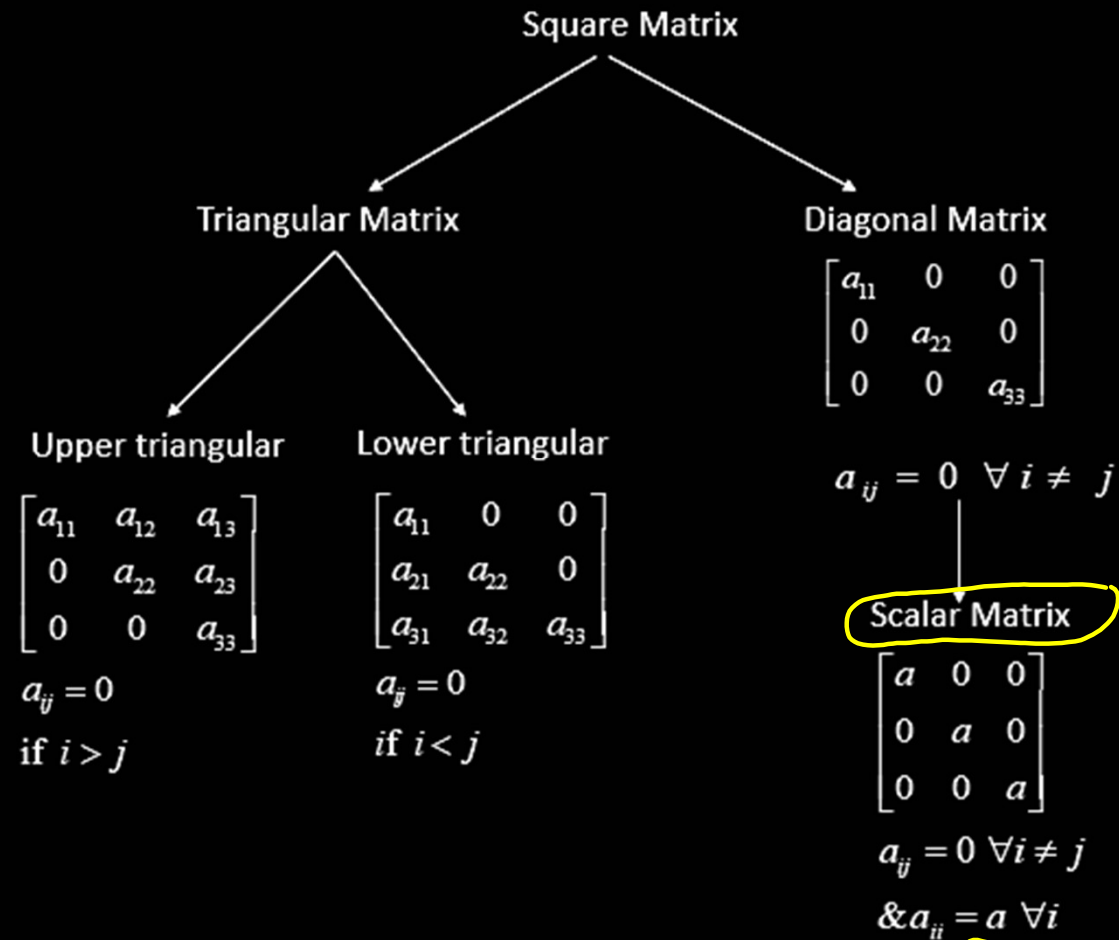
Upper Triangular
 $a_{ij} = 0 \forall i > j$



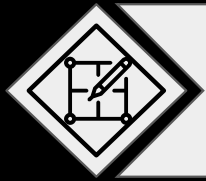
Lower Triangular
 $a_{ij} = 0 \forall i < j$



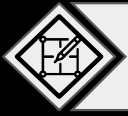
$a_{ij} = 0 \forall i \neq j$



In particular scalar matrix having 1 at diagonal is called identity or unit matrix.
(Denoted by $I_{n \times n}$).



Algebra of Matrices



Algebra of Matrices

Equality of matrices

Two matrices are said to be equal if

1. They have same order
2. Corresponding terms are equal.



For what values of x and y are the following matrices equal ?

$$A = \begin{bmatrix} 2x+1 & 3y \\ 0 & y^2-5y \end{bmatrix}_{2 \times 2}, B = \begin{bmatrix} x+3 & y^2+2 \\ 0 & -6 \end{bmatrix}_{2 \times 2}$$

Solution:

$$2x+1 = x+3 \quad \& \quad 3y = y^2+2 \quad \& \quad y^2-5y = -6$$

$x=2$ $y=1, 2$ $y=2, 3$

$y=2$

Ans: $x=2$
 $y=2$

Note

Q

Solve for real x:

$$\begin{bmatrix} x^4 - 2x^3 + 3x^2 \\ 2 \end{bmatrix}_{2 \times 1} = \begin{bmatrix} 2x - 1 \\ 2 \end{bmatrix}_{2 \times 1}$$

Sol: $x^4 - 2x^3 + 3x^2 - 2x + 1 = 0$

$$x^2 - 2x + 3 - \frac{2}{x} + \frac{1}{x^2} = 0$$

$$\left(x^2 + \frac{1}{x^2}\right) - 2\left(x + \frac{1}{x}\right) + 3 = 0$$

Put $x + \frac{1}{x} = t$

$$(t^2 - 2) - 2t + 3 = 0$$

$$t^2 - 2t + 1 = 0 \Rightarrow (t - 1) \Rightarrow x + \frac{1}{x} = 1 \Rightarrow x^2 - x + 1 = 0$$

No real soln

Recall

(1) If $x + \frac{1}{x} = t$ then $x^2 + \frac{1}{x^2} = t^2 - 2$

(2) If $x - \frac{1}{x} = t$ then $x^2 + \frac{1}{x^2} = t^2 + 2$



Solve for real x:

$$\begin{bmatrix} x^4 - 2x^3 + 3x^2 \\ 2 \end{bmatrix}_{2 \times 1} = \begin{bmatrix} 2x - 1 \\ 2 \end{bmatrix}_{2 \times 1}$$

Solution:

On comparing both sides of given equation, we get

$$x^4 - 2x^3 + 3x^2 = 2x - 1$$

Dividing by x^2 , we get

$$x^2 - 2x + 3 - \frac{2}{x} + \frac{1}{x^2} = 0$$

$$\Rightarrow x^2 + \frac{1}{x^2} - 2\left(x + \frac{1}{x}\right) + 3 = 0$$

$$\Rightarrow \left(x + \frac{1}{x}\right)^2 - 2 - 2\left(x + \frac{1}{x}\right) + 3 = 0$$



Solve for real x:

$$\begin{bmatrix} x^4 - 2x^3 + 3x^2 \\ 2 \end{bmatrix}_{2 \times 1} = \begin{bmatrix} 2x - 1 \\ 2 \end{bmatrix}_{2 \times 1}$$

Solution:

$$\text{Let } x + \frac{1}{x} = t$$

$$\Rightarrow t^2 - 2t + 1 = 0$$

$$\Rightarrow (t - 1)^2 = 0 \quad \therefore t = 1$$

$$\Rightarrow x + \frac{1}{x} = 1$$

$$\Rightarrow x^2 + 1 - x = 0$$

$\therefore$ No value of x



Find the relation between p , q and r for which the following matrices are equal:

$$A = \begin{bmatrix} 4p^2 + 16q^2 + r^2 \\ 1 \end{bmatrix}_{2 \times 1}, \quad B = \begin{bmatrix} 8pq + 4qr + 2rp \\ 1 \end{bmatrix}_{2 \times 1}$$

$$4p^2 + 16q^2 + r^2 = 8pq + 4qr + 2pr$$

$$\left[\begin{array}{l} \text{Recall} \\ x^2 + y^2 + z^2 - xy - yz - zx = 0 \\ \Rightarrow x = y = z \end{array} \right]$$

$$(2p)^2 + (4q)^2 + (r)^2 - (2p)(4q) - (4q)(r) - (2p)(r) = 0$$

$$\boxed{2p = 4q = r}$$



Find the relation between p, q and r for which the following matrices are equal:

$$A = \begin{bmatrix} 4p^2 + 16q^2 + r^2 \\ 1 \end{bmatrix}_{2 \times 1}, \quad B = \begin{bmatrix} 8pq + 4qr + 2rp \\ 1 \end{bmatrix}_{2 \times 1}$$

Solution:

$$A = \begin{bmatrix} 4p^2 + 16q^2 + r^2 \\ 1 \end{bmatrix}_{2 \times 1}, \quad B = \begin{bmatrix} 8pq + 4qr + 2rp \\ 1 \end{bmatrix}_{2 \times 1}$$

Since both matrices are equal, on equating we get

$$4p^2 + 16q^2 + r^2 - 8pq - 4qr - 2rp = 0$$

$$\Rightarrow (2p)^2 + (4q)^2 + (r)^2 - (2p)(4q) - (4q)(r) - (2p)(r) = 0$$

Using property

$$\text{If } a^2 + b^2 + c^2 - ab - bc - ca = 0 \Rightarrow a = b = c$$

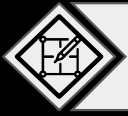
$$\Rightarrow 2p = 4q = r$$



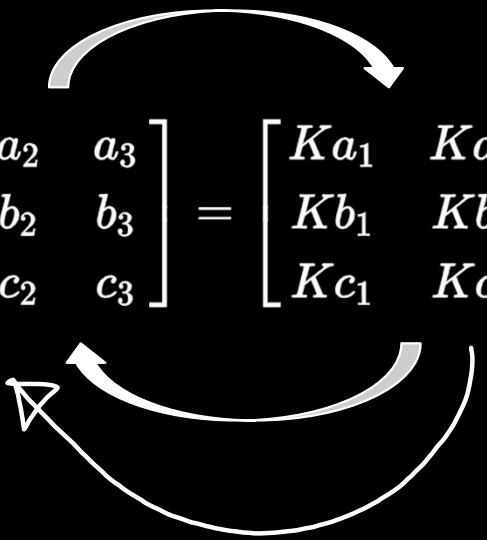
Algebra of Matrices

Two matrices of same order are added term by term

$$\begin{bmatrix} \textcircled{2} & \textcircled{3} & -4 \\ 5 & 0 & 1 \end{bmatrix}_{2 \times 3} + \begin{bmatrix} \textcircled{5} & \textcircled{1} & 5 \\ 0 & 2 & 1 \end{bmatrix}_{2 \times 3} = \begin{bmatrix} \textcircled{7} & \textcircled{4} & 1 \\ 5 & 2 & 2 \end{bmatrix}$$



Algebra of Matrices

$$K \times \begin{bmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{bmatrix} = \begin{bmatrix} Ka_1 & Ka_2 & Ka_3 \\ Kb_1 & Kb_2 & Kb_3 \\ Kc_1 & Kc_2 & Kc_3 \end{bmatrix}$$




If $A = \begin{bmatrix} 5 & -1 \\ 3 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} -4 & 3 \\ 1 & -2 \end{bmatrix}$, find the matrix X such that $3A - 2B + 3X = 0$

$$3A - 2B + 3X = 0$$

$$3X = -3A + 2B$$

$$X = -A + \frac{2}{3}B$$

$$= \begin{bmatrix} -5 & 1 \\ -3 & -2 \end{bmatrix} + \begin{bmatrix} -\frac{8}{3} & \frac{4}{3} \\ \frac{2}{3} & -\frac{4}{3} \end{bmatrix}$$

=



If $A = \begin{bmatrix} 5 & -1 \\ 3 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} -4 & 3 \\ 1 & -2 \end{bmatrix}$, find the matrix X such that $3A - 2B + 3X = 0$

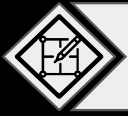
Solution:

We have, $3A - 2B + 3X = 0 \Rightarrow 3X = 2B - 3A$

$$\begin{aligned} \Rightarrow 3X &= 2 \begin{bmatrix} -4 & 3 \\ 1 & -2 \end{bmatrix} - 3 \begin{bmatrix} 5 & -1 \\ 3 & 2 \end{bmatrix} \\ &= \begin{bmatrix} -8 - 15 & 6 + 3 \\ 2 - 9 & -4 - 6 \end{bmatrix} = \begin{bmatrix} -23 & 9 \\ -7 & -10 \end{bmatrix} \end{aligned}$$

$$\Rightarrow X = \frac{1}{3} \begin{bmatrix} -23 & 9 \\ -7 & -10 \end{bmatrix}$$

$$\swarrow = \begin{bmatrix} -\frac{23}{3} & 3 \\ -\frac{7}{3} & -\frac{10}{3} \end{bmatrix}$$

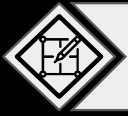


Algebra of Matrices

Multiplication of two Matrices

$$\underline{A}_{m \times n} \times \underline{B}_{n \times p} = C_{m \times p}$$

same



Algebra of Matrices

Multiplication of two Matrices

$$A_{m \times n} \times B_{n \times p} = C_{m \times p}$$

Diagram illustrating the multiplication of two matrices:

- $A_{m \times n}$ is labeled as the **Pre-multiplier**.
- $B_{n \times p}$ is labeled as the **Post multiplier**.



Algebra of Matrices

Multiplication of two Matrices

$$A_{m \times n} \times B_{n \times p} = C_{m \times p}$$

Diagram illustrating the multiplication of two matrices:

- $A_{m \times n}$ is labeled as the **Pre-multiplier**.
- $B_{n \times p}$ is labeled as the **Post multiplier**.



NOTE

- Two matrices can be multiplied only if number of columns of pre-multiplier is equal to number of rows of post multiplier.
- Matrices are multiplied in row by column way of determinant multiplications.



Evaluate the following:

$$(a) \begin{bmatrix} 2 & 3 & 4 \\ 1 & 2 & 3 \end{bmatrix}_{2 \times 3} \times \begin{bmatrix} 1 & 0 & 1 & 2 \\ 2 & 1 & 1 & 2 \\ 0 & 1 & -1 & 3 \end{bmatrix}_{3 \times 4}$$

$$(b) [1 \ 0 \ 2 \ 3]_{1 \times 4} \times \begin{bmatrix} 1 & 1 \\ 2 & 2 \\ 3 & 0 \\ 4 & 1 \end{bmatrix}_{4 \times 2}$$

$$(c) \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}_{3 \times 1} \times [1 \ 2 \ 4]_{1 \times 3}$$

Q

Evaluate the following:

$$(a) \begin{bmatrix} 2 & 3 & 4 \\ 1 & 2 & 3 \end{bmatrix}_{2 \times 3} \times \begin{bmatrix} 1 & 0 & 1 & 2 \\ 2 & 1 & 1 & 2 \\ 0 & 1 & -1 & 3 \end{bmatrix}_{3 \times 4}$$

Solution:

$$\begin{bmatrix} 8 & 7 & 1 & 22 \\ 5 & 5 & 0 & 15 \end{bmatrix}_{2 \times 4}$$

→ C_{23}

2nd row
of pre-multiplier

3rd colⁿ of
post multiplier.

Product ka
 ij^{th} element
Kaise aata hai?



Evaluate the following:

$$(b) [1 \ 0 \ 2 \ 3]_{1 \times 1} \times \begin{bmatrix} 1 & 1 \\ 2 & 2 \\ 3 & 0 \\ 4 & 1 \end{bmatrix}_{1 \times 2}$$

Solution:

$$[18 \ 4]_{1 \times 2}$$

Q

Evaluate the following:

$$(c) \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}_{3 \times 1} \times [1 \ 2 \ 4]_{1 \times 3}$$

Solution:

$$\begin{array}{l} \rightarrow \\ \rightarrow \\ \rightarrow \end{array} \begin{bmatrix} 1 & 2 & 4 \\ 1 & 2 & 4 \\ 0 & 0 & 0 \end{bmatrix}_{3 \times 3}$$



If A & B are two matrices of order 2 such that

$$A + B = \begin{bmatrix} 2 & 0 \\ 3 & 4 \end{bmatrix} \text{ and } A - B = \begin{bmatrix} 0 & 2 \\ 5 & -2 \end{bmatrix}, \text{ then find } A^3.$$

$$\begin{aligned} & \text{②} \quad \text{③} \quad \text{④} \quad \text{⑤} \quad \text{⑥} \quad \text{⑦} \quad \text{⑧} \quad \text{⑨} \quad \text{⑩} \quad \text{⑪} \quad \text{⑫} \quad \text{⑬} \quad \text{⑭} \quad \text{⑮} \quad \text{⑯} \quad \text{⑰} \quad \text{⑱} \quad \text{⑲} \quad \text{⑳} \quad \text{\textcircled{+}} \\ 2A &= \begin{bmatrix} 2 & 2 \\ 8 & 2 \end{bmatrix} \\ A &= \begin{bmatrix} 1 & 1 \\ 4 & 1 \end{bmatrix} \end{aligned}$$



If A & B are two matrices of order 2 such that

$$A + B = \begin{bmatrix} 2 & 0 \\ 3 & 4 \end{bmatrix} \text{ and } A - B = \begin{bmatrix} 0 & 2 \\ 5 & -2 \end{bmatrix}, \text{ then find } A^3.$$

Solution:

$$A + B = \begin{bmatrix} 2 & 0 \\ 3 & 4 \end{bmatrix}, \quad A - B = \begin{bmatrix} 0 & 2 \\ 5 & -2 \end{bmatrix}$$

$$A + B + A - B = 2A$$

$$\Rightarrow \begin{bmatrix} 2+0 & 0+2 \\ 3+5 & 4-2 \end{bmatrix} = 2A$$

$$\Rightarrow A = \begin{bmatrix} \frac{2}{2} & \frac{2}{2} \\ \frac{8}{2} & \frac{2}{2} \end{bmatrix}$$

$$\Rightarrow A = \begin{bmatrix} 1 & 1 \\ 4 & 1 \end{bmatrix}$$



If A & B are two matrices of order 2 such that

$$A + B = \begin{bmatrix} 2 & 0 \\ 3 & 4 \end{bmatrix} \text{ and } A - B = \begin{bmatrix} 0 & 2 \\ 5 & -2 \end{bmatrix}, \text{ then find } A^3.$$

Solution:

$$A^3 = A^2 \cdot A$$

$$A^2 = A \cdot A = \begin{bmatrix} 1 & 1 \\ 4 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 1 \\ 4 & 1 \end{bmatrix}$$

$$A^2 = \begin{bmatrix} 1+4 & 1+1 \\ 4+4 & 4+1 \end{bmatrix} = \begin{bmatrix} 5 & 2 \\ 8 & 5 \end{bmatrix}$$

$$A^3 = \begin{bmatrix} 5 & 2 \\ 8 & 5 \end{bmatrix} \cdot \begin{bmatrix} 1 & 1 \\ 4 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 5+10 & 5+2 \\ 8+20 & 8+5 \end{bmatrix}$$

$$= \begin{bmatrix} 15 & 7 \\ 28 & 13 \end{bmatrix}$$



Algebra of Matrices

Properties of Multiplication

1. It's not commutative.

i.e. $AB \neq BA$ (in general)

Remr

$$\left. \begin{array}{l} AB = A \\ \& BA = B \end{array} \right\} \text{then} \begin{array}{l} A^2 = A \\ B^2 = B \end{array}$$



If $\underline{AB = A}$ and $\underline{BA = B}$ then $B^2 =$

$$B \underline{AB} = B^2$$

$$B A = B^2$$
$$\rightarrow B = B^2$$

(A) A^2

(B) B

(C) A

(D) $-B$



If $AB = A$ and $BA = B$ then $B^2 =$

☐ A A^2

☒ B B

☐ C A

☐ D $-B$



If $A^2 = A$ then show that: $(I + A)^3 = I + 7A$

$$A^3 = A^2 \cdot A = A$$

$$\begin{aligned} (I + A)^3 &= I^3 + A^3 + 3I^2A + 3A^2I \\ &= I + A + 3A + 3A \\ &= I + 7A \end{aligned}$$

$$IA = AI = A$$

I and A commute with each other: $I \times A = A \times I = A$



If $A^2 = A$ then show that: $(I + A)^3 = I + 7A$

Solution:

Given, $A^2 = A$

Then

$$(I + A)^3 = I^3 + A^3 + 3I^2A + 3IA^2$$

$$= I + A + 3A + 3A$$

$$= I + 7A$$

Single correct question

JEE Advanced 2022, P2

Q

If $M = \begin{pmatrix} \frac{5}{2} & \frac{3}{2} \\ -\frac{3}{2} & -\frac{1}{2} \end{pmatrix}$, then which of the following matrices is equal to M^{2022} ?

$$M = \begin{bmatrix} 5/2 & 3/2 \\ -3/2 & -1/2 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} 3/2 & 3/2 \\ -3/2 & -3/2 \end{bmatrix}$$

$$= I + \frac{3}{2} \begin{bmatrix} 1 & 1 \\ -1 & -1 \end{bmatrix} \quad A$$

$$M = I + \frac{3}{2} A$$

$$M^{2022} = \left(I + \frac{3}{2} A \right)^{2022}$$

$$= {}^{2022}C_0 \times I^{2022} + {}^{2022}C_1 I^{2021} \times \frac{3}{2} A + 0$$

(A) $\begin{pmatrix} 3034 & 3033 \\ -3033 & -3032 \end{pmatrix}$

(B) $\begin{pmatrix} 3034 & -3033 \\ 3033 & -3032 \end{pmatrix}$

(C) $\begin{pmatrix} 3033 & 3032 \\ -3032 & -3031 \end{pmatrix}$

(D) $\begin{pmatrix} 3032 & 3031 \\ -3031 & -3030 \end{pmatrix}$

$$\left(A^2 = \begin{bmatrix} 1 & 1 \\ -1 & -1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ -1 & -1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \right)$$

Single correct question

JEE Advanced 2022, P2



If $M = \begin{pmatrix} \frac{5}{2} & \frac{3}{2} \\ -\frac{3}{2} & -\frac{1}{2} \end{pmatrix}$, then which of the following matrices is equal to M^{2022} ?

$$\begin{aligned} \text{Ans} &= I + 2022 \times A \times \frac{3}{2} \\ &= I + 2022 \times \frac{3}{2} \begin{bmatrix} 1 & 1 \\ -1 & -1 \end{bmatrix} \end{aligned}$$

(A) $\begin{pmatrix} 3034 & 3033 \\ -3033 & -3032 \end{pmatrix}$

(B) $\begin{pmatrix} 3034 & -3033 \\ 3033 & -3032 \end{pmatrix}$

(C) $\begin{pmatrix} 3033 & 3032 \\ -3032 & -3031 \end{pmatrix}$

(D) $\begin{pmatrix} 3032 & 3031 \\ -3031 & -3030 \end{pmatrix}$

Single correct question

JEE Advanced 2022, P2



If $M = \begin{pmatrix} \frac{5}{2} & \frac{3}{2} \\ -\frac{3}{2} & -\frac{1}{2} \end{pmatrix}$, then which of the following matrices is equal to M^{2022} ?

Solution:

$$M = \begin{bmatrix} \frac{5}{2} & \frac{3}{2} \\ -\frac{3}{2} & -\frac{1}{2} \end{bmatrix} = I + \frac{3}{2} \begin{bmatrix} 1 & 1 \\ -1 & -1 \end{bmatrix}$$

$$A^2 = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}, \text{ where } A = \begin{bmatrix} 1 & 1 \\ -1 & -1 \end{bmatrix}$$

$$\begin{aligned} \Rightarrow M^{2022} &= \left(I + \frac{3}{2} A \right)^{2022} = I + \frac{3}{2} \times 2022 A \\ &= \begin{bmatrix} 3034 & 3033 \\ -3033 & -3032 \end{bmatrix} \end{aligned}$$

complete it

Q

Single correct question

JEE Advanced 2016, P2

Let $P = \begin{bmatrix} 1 & 0 & 0 \\ 4 & 1 & 0 \\ 16 & 4 & 1 \end{bmatrix}$ and I be the identity matrix of order 3.

If $Q = [q_{ij}]$ is a matrix such that $P^{50} - Q = I$, then $\frac{q_{31} + q_{32}}{q_{21}} =$

$$P^{50} - Q = I$$

$$Q = P^{50} - I$$

$$P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} + 4 \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 4 & 1 & 0 \end{bmatrix}$$

$$P = I + 4A \Rightarrow P^{50} = (I + 4A)^{50} = {}^{50}C_0 I + {}^{50}C_1 I \times (4A)^1 + {}^{50}C_2 I (4A)^2 + 0$$

$\underbrace{\quad}_{A^3=0}$

(A) 52

(B) 103

(C) 201

(D) 205

Single correct question

JEE Advanced 2016, P2



Let $P = \begin{bmatrix} 1 & 0 & 0 \\ 4 & 1 & 0 \\ 16 & 4 & 1 \end{bmatrix}$ and I be the identity matrix of order 3.

If $Q = [q_{ij}]$ is a matrix such that $P^{50} - Q = I$, then $\frac{q_{31} + q_{32}}{q_{21}} =$

(A) 52

(B) 103

(C) 201

(D) 205

Single correct question

JEE Advanced 2016, P2



Let $P = \begin{bmatrix} 1 & 0 & 0 \\ 4 & 1 & 0 \\ 16 & 4 & 1 \end{bmatrix}$ and I be the identity matrix of order 3.

If $Q = [q_{ij}]$ is a matrix such that $P^{50} - Q = I$, then $\frac{q_{31} + q_{32}}{q_{21}} =$

Solution:

$$P = \begin{bmatrix} 1 & 0 & 0 \\ 4 & 1 & 0 \\ 16 & 4 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 4 & 0 & 0 \\ 16 & 4 & 0 \end{bmatrix} + \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\text{let } A = \begin{bmatrix} 0 & 0 & 0 \\ 4 & 0 & 0 \\ 16 & 4 & 0 \end{bmatrix} \Rightarrow A^2 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 16 & 0 & 0 \end{bmatrix}$$

$$\text{and } A^3 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Single correct question

JEE Advanced 2016, P2



Let $P = \begin{bmatrix} 1 & 0 & 0 \\ 4 & 1 & 0 \\ 16 & 4 & 1 \end{bmatrix}$ and I be the identity matrix of order 3.

If $Q = [q_{ij}]$ is a matrix such that $P^{50} - Q = I$, then $\frac{q_{31} + q_{32}}{q_{21}} =$

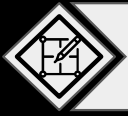
Solution:

$\Rightarrow A^n$ is a null matrix $\forall n \geq 3$

$$P^{50} = (I + A)^{50} = I + 50A + \frac{50 \times 49}{2} A^2$$

$$Q + I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} + 50 \begin{bmatrix} 0 & 0 & 0 \\ 4 & 0 & 0 \\ 16 & 4 & 0 \end{bmatrix} + 25 \times 49 \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 16 & 0 & 0 \end{bmatrix}$$

$$\begin{aligned} \Rightarrow \left(\frac{q_{31} + q_{32}}{q_{21}} \right) &= \frac{16(50 + 25 \times 49) + 50 \times 4}{50 \times 4} \\ &= \frac{16 \times 51 + 8}{8} = 102 + 1 = 103 \end{aligned}$$



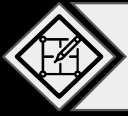
Algebra of Matrices

Properties of Multiplication

2. Its Associative

i.e. $(A \times B) \times C = A \times (B \times C)$

same $\begin{matrix} \nearrow (A \times B) \times C \\ \searrow A \times (B \times C) \end{matrix}$



Algebra of Matrices

Properties of Multiplication

3. It distributes over addition.

$$\text{i.e. } A \times (B + C) = A \times B + A \times C$$

$$\text{or } (B + C) \times A = B \times A + C \times A$$

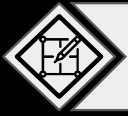


Algebra of Matrices

Properties of Multiplication

4. Multiplicative identity exists.

$$I_{m \times m} \times A_{m \times n} = A_{m \times n} = A_{m \times n} \times I_{n \times n}$$



Algebra of Matrices

Properties of Multiplication

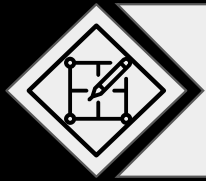
5. Any matrix multiplied with null matrix gives a null matrix.

$$\underline{\underline{A}} \times \underline{\underline{B}} = \underline{\underline{O}} \not\Rightarrow A = \underline{\underline{O}} \text{ or } B = \underline{\underline{O}}$$

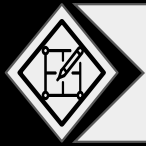


Remark

If $A \times B$ is null matrix then it is not necessary that either A or B will be null matrix.



Relation between Matrices & Determinants



Relation between Matrices & Determinants

(a) We find determinant of a square matrix.



Relation between Matrices & Determinants

(a) We find determinant of a square matrix.

(b) If A & B are two square matrices of same order then $\underline{|A \times B| = |A| \times |B|}$

$$|A \times A| = |A| \times |A|$$

$$|A^2| = |A|^2$$



Remark

Clearly, $\underline{|A^n| = |A|^n}$



Relation between Matrices & Determinants

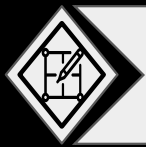
- (a) We find determinant of a square matrix.
- (b) If A & B are two square matrices of same order then $|A \times B| = |A| \times |B|$
- (c) If A_n is a square matrix of order n & K is a constant then:

$$|K \times A_n| = K^n |A_n|$$



Remark

Clearly, $|A^n| = |A|^n$



Relation between Matrices & Determinants

(c) If A_n is a square matrix of order n & K is a constant then:

$$|K \times A_n| = K^n |A_n|$$

Consider $A_3 = \begin{bmatrix} 1 & 2 & 3 \\ 2 & -2 & 0 \\ 3 & 5 & 1 \end{bmatrix}$

$$\therefore K \times A_3 = \begin{bmatrix} K & 2K & 3K \\ 2K & -2K & 0 \\ 3K & 5K & K \end{bmatrix}$$



Relation between Matrices & Determinants

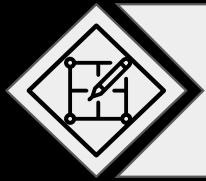
(c) If A_n is a square matrix of order n & K is a constant then:

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$$\therefore K \times A_3 = \begin{bmatrix} K & 2K & 3K \\ 2K & -2K & 0 \\ 3K & 5K & K \end{bmatrix}$$

$$|K \times A_3| = \begin{vmatrix} K & 2K & 3K \\ 2K & -2K & 0 \\ 3K & 5K & K \end{vmatrix} = K^3 \times \begin{vmatrix} 1 & 2 & 3 \\ 2 & -2 & 0 \\ 3 & 5 & 1 \end{vmatrix} = K^3 \times |A_3|$$



Transpose and its properties



Transpose and its properties

Matrix obtained by interchanging rows & columns is called transpose of matrix, denoted by A^T or A' .

Properties of transpose

1. $(A^T)^T = A$
2. $(A + B)^T = A^T + B^T$
3. $(KA)^T = K(A^T)$; K is Constant
4. $(AB)^T = B^T A^T$



Transpose and its properties

Matrix obtained by interchanging rows & columns is called transpose of matrix, denoted by A^T or A' .

Properties of transpose

1. $(A^T)^T = A$
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3. $(KA)^T = K(A^T)$; K is Constant
4. $(AB)^T = B^T A^T$



Remark

1. $(ABC)^T = C^T B^T A^T$
2. $(A^n)^T = (A^T)^n$



A square matrix P satisfies $P^2 = I - P$, where I is an identity matrix. If $P^n = 5I - 8P$, then $n =$

Sol. $P^2 = I - P$ $\Rightarrow I - P$

$$P^3 = I + P^2 - 2P$$

$$P^3 = 2I - 3P$$

$$P^4 = 2P - 3P^2 \Rightarrow I - P$$

$$P^5 = 2I - 3I + 3P$$

$$P^5 = 5P - 3I \Rightarrow P^6 = 5P^2 - 3I$$

(A) 4

(B) 5

(C) 6

(D) 7



A square matrix P satisfies $P^2 = I - P$, where I is an identity matrix. If $P^n = 5I - 8P$, then $n =$

A 4

B 5

C 6

D 7



A square matrix P satisfies $P^2 = I - P$, where I is an identity matrix. If $P^n = 5I - 8P$, then $n =$

Solution:

$$\text{Since } P^2 = I - P \text{ (Given) } \dots\dots\dots (1)$$

$$P^3 = P(1 - P)$$

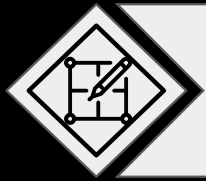
$$P^3 = P - P^2 = P - (I - P) \text{ (Using) } \dots\dots\dots (2)$$

$$P^3 = 2P - I$$

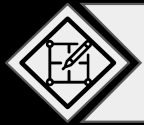
$$\text{Similarly } P^4 = 2P^2 - P = 2I - 3P \text{ and } P^5 = 5P - 3I$$

$$P^6 = 5P^2 - 3P = 5I - 8P$$

$$\therefore n = 6$$



Some special types of Matrices



Some special types of Matrices

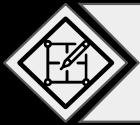
Symmetric and Skew symmetric Matrix

Symmetric matrix

If $A_n^T = A_n$ then square matrix A_n is called symmetric matrix.

i.e. $a_{ij} = a_{ji} \forall i \& j$

$$\begin{bmatrix} 2 & 1 & 3 \\ 1 & 0 & -2 \\ 3 & -2 & 5 \end{bmatrix}$$



Some special types of Matrices

Symmetric and Skew symmetric Matrix

Symmetric matrix

If $A_n^T = A_n$ then square matrix A_n is called symmetric matrix.

i.e. $a_{ij} = a_{ji} \forall i \& j$

$$\begin{bmatrix} 2 & 1 & 3 \\ 1 & 0 & -2 \\ 3 & -2 & 5 \end{bmatrix}$$

Skew Symmetric matrix

If $A_n^T = -A_n$ then square matrix A_n is called skew symmetric matrix.

i.e. $a_{ij} = -a_{ji} \forall i \& j$

Clearly, $a_{ii} = 0 \forall i$

$$\begin{bmatrix} 0 & 2 & 4 \\ -2 & 0 & -3 \\ -4 & 3 & 0 \end{bmatrix}$$

$$\begin{aligned} a_{ii} &= -a_{ii} \\ 2a_{ii} &= 0 \\ a_{ii} &= 0 \end{aligned}$$



If A is a square matrix then prove that:

(a) $\frac{1}{2}(A + A^T)$ is a symmetric matrix

(b) $\frac{1}{2}(A - A^T)$ is a skew symmetric matrix



If A is a square matrix then prove that:

(a) $\frac{1}{2}(A + A^T)$ is a symmetric matrix

Solution:

$$\text{Let } C = \frac{1}{2}(A + A^T)$$

$$\begin{aligned} C^T &= \left[\frac{1}{2}(A + A^T) \right]^T \\ &= \frac{1}{2}(A^T + (A^T)^T) \\ &= \frac{1}{2}(A^T + A) = C \end{aligned}$$



If A is a square matrix then prove that:

(b) $\frac{1}{2}(A - A^T)$ is a skew symmetric matrix

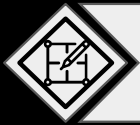
Solution:

$$\text{Let } C = \frac{1}{2}(A - A^T)$$

$$C^T = \frac{1}{2}(A^T - (A^T)^T)$$

$$= \frac{1}{2}(A^T - A)$$

$$= -\frac{1}{2}(A - A^T) = -C$$



Some special types of Matrices

Symmetric and Skew symmetric Matrix



Remark

Every square matrix A can be represented as a sum of symmetric & skew symmetric matrix.

$$A = \frac{1}{2}(A + A^T) + \frac{1}{2}(A - A^T)$$

$$\begin{aligned} A^T &= A \\ B^T &= B \end{aligned}$$



If A and B are symmetric matrices of same order then prove that AB - BA is skew symmetric matrix.

$$\text{Let } C = AB - BA$$

$$\begin{aligned} C^T &= (AB - BA)^T \\ &= (AB)^T - (BA)^T \\ &= B^T A^T - A^T B^T \\ &= BA - AB \\ &= -C. \end{aligned}$$



If A and B are symmetric matrices of same order then prove that $AB - BA$ is skew symmetric matrix.

Solution:

It is given that A and B are symmetric matrices.

Therefore, we have:

$$A' = A \text{ and } B' = B \quad \text{..... (1)}$$

Now,

$$\begin{aligned} (AB - BA)' &= (AB)' - (BA)' && [\because (A - B)' = A' - B'] \\ &= B'A' - A'B' && [\because (AB)' = B'A'] \\ &= BA - AB && \text{..... [using (1)]} \\ &= -(AB - BA) \end{aligned}$$

Thus, $(AB - BA)$ is a skew-symmetric matrix.

Multiple correct question

JEE Advanced 2015



Let $X_{3 \times 3}$ & $Y_{3 \times 3}$ are non zero skew symmetric matrices & $Z_{3 \times 3}$ is a non-zero symmetric matrix then which of the following are skew symmetric.

Sol

$$\begin{aligned} X^T &= -X \\ Y^T &= -Y \\ Z^T &= Z \end{aligned}$$

$$\begin{aligned} \text{(a) Let } C &= Y^3 Z^4 - Z^4 Y^3 \\ C^T &= (Y^3 Z^4 - Z^4 Y^3)^T \\ &= (Y^3 Z^4)^T - (Z^4 Y^3)^T \\ &= (Z^T)^4 (Y^T)^3 - (Y^T)^3 (Z^T)^4 \\ &= -Z^4 Y^3 + Y^3 Z^4 \\ &= Y^3 Z^4 - Z^4 Y^3 \\ &= C \end{aligned}$$

Sym (A) $Y^3 Z^4 - Z^4 Y^3$

Sym (B) $X^{44} + Y^{44}$

(C) $X^4 Z^3 - Z^3 X^4$

(D) $X^{23} + Y^{23}$

Multiple correct question

JEE Advanced 2015



Let $X_{3 \times 3}$ & $Y_{3 \times 3}$ are non zero skew symmetric matrices & $Z_{3 \times 3}$ is a non-zero symmetric matrix then which of the following are skew symmetric.

- (A) $Y^3Z^4 - Z^4Y^3$
- (B) $X^{44} + Y^{44}$
- (C) $X^4Z^3 - Z^3X^4$
- (D) $X^{23} + Y^{23}$

Multiple correct question

JEE Advanced 2015



Let $X_{3 \times 3}$ & $Y_{3 \times 3}$ are non zero skew symmetric matrices & $Z_{3 \times 3}$ is a non-zero symmetric matrix then which of the following are skew symmetric.

Solution:

Given X and Y are skew symmetric

Then, $X^T = -X$ and $Y^T = -Y$

Also, $Z^T = Z$

Now, in (c) option,

$$\begin{aligned}(X^4 Z^3 - Z^3 X^4)^T &= (X^4 Z^3)^T - (Z^3 X^4)^T \\&= (Z^T)^3 (X^T)^4 - (X^T)^4 (Z^T)^3 \\&= Z^3 X^4 - X^4 Z^3 \\&= -(X^4 Z^3 - Z^3 X^4)\end{aligned}$$

Multiple correct question

JEE Advanced 2015



Let $X_{3 \times 3}$ & $Y_{3 \times 3}$ are non zero skew symmetric matrices & $Z_{3 \times 3}$ is a non-zero symmetric matrix then which of the following are skew symmetric.

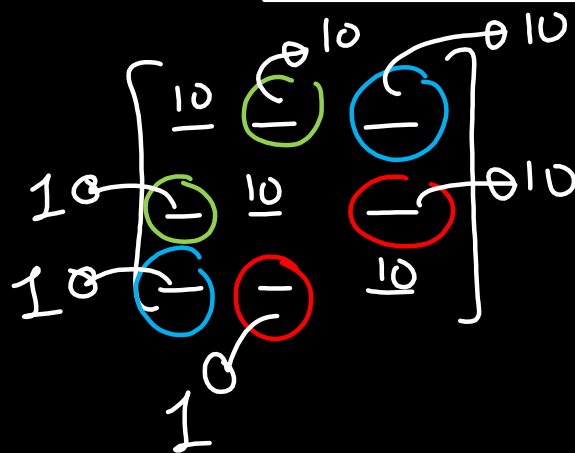
Solution:

$$\begin{aligned} (X^{23} + Y^{23})^T &= -X^{23} - Y^{23} \\ &= -(X^{23} + Y^{23}) \end{aligned}$$

Thus both are skew symmetric.



The number of symmetric matrices of order 3, with all the entries from the set $\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$ is:



(A) 10^9

(B) 9^{10}

(C) 10^6

(D) 6^{10}

JEE Main 13th April, 2023



The number of symmetric matrices of order 3, with all the entries from the set $\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$ is:

- ☐ A 10^9
- ☐ B 9^{10}
- ☒ C 10^6
- ☐ D 6^{10}



The number of symmetric matrices of order 3, with all the entries from the set $\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$ is:

Solution:

A 3 x 3 symmetric matrix looks like this :

$$\begin{bmatrix} a & b & c \\ b & d & e \\ c & e & f \end{bmatrix}$$

Notice that there are only 6 unique elements, we need to fill because of the symmetry.

Each of these unique elements can take a value from the set $\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$, which has 10 elements.

So, the total number of symmetric matrices can be calculated as $10 \times 10 \times 10 \times 10 \times 10 \times 10 = 10^6$



Some special types of Matrices

Symmetric and Skew symmetric Matrix

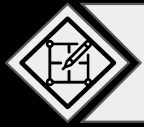


NOTE

Determinant of skew symmetric matrix of odd order is zero.

$$\left. \begin{array}{l} A_n^T = -A_n, \text{ } n\text{-odd} \\ \Rightarrow |A| = 0 \end{array} \right\} \text{Pf:}$$

$$\begin{aligned} A_n^T &= -A_n \\ |A_n^T| &= |-A_n| \\ |A_n| &= (-1)^n |A_n| \\ &\quad \swarrow \text{ } n\text{-odd} \\ 2|A| &= 0 \end{aligned} \quad \left(|K \times A_n| = K^n |A_n| \right)$$



Some special types of Matrices

Hermitian and Skew Hermitian Matrix



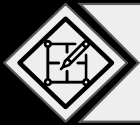
Some special types of Matrices

Hermitian and Skew Hermitian Matrix

Transposed Conjugate: Transpose of conjugate of a matrix is called its transposed conjugate. Denoted by $(\bar{A})^T$ or A^θ

If $A = \begin{bmatrix} i & 2+i & 1 \\ 0 & 1-i & 1 \\ 2 & 1+2i & 0 \end{bmatrix}$ then

$\bar{A} = \begin{bmatrix} -i & 2-i & 1 \\ 0 & 1+i & 1 \\ 2 & 1-2i & 0 \end{bmatrix}$ and $(\bar{A})^T = \begin{bmatrix} -i & 0 & 2 \\ 2-i & 1+i & 1 \\ 1 & 1 & 0 \end{bmatrix}$



Some special types of Matrices

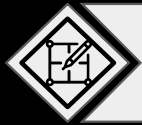
Hermitian and Skew Hermitian Matrix

Hermitian Matrix: A square matrix is called Hermitian matrix if $(\bar{A})^T = A$

i.e., $\bar{a}_{ij} = a_{ji} \forall i, j \Rightarrow (\bar{A})^T = A$

Clearly, $\bar{a}_{ii} = a_{ii}$ and hence diagonal elements will be purely real.

$$\begin{bmatrix} 1 & 2+i & 1 \\ 2-i & 2 & 1+2i \\ 1 & 1-2i & 0 \end{bmatrix} = A$$



Some special types of Matrices

Hermitian and Skew Hermitian Matrix

Skew Hermitian Matrix: A square matrix is called Skew Hermitian matrix

if $(\overline{A})^T = -A$

i.e., $\overline{a_{ji}} = -a_{ij} \forall i, j$

$\Rightarrow \overline{a_{ii}} = -a_{ii}$
 $\Rightarrow a_{ii} \rightarrow \text{purely img.}$

Clearly, $\overline{a_{ii}} = -a_{ii} \forall i$ and hence diagonal elements will be purely imaginary.

$$\begin{bmatrix} i & 1+i & 1+2i \\ -(1-i) & 2i & 2 \\ -(1-2i) & 2 & 0 \end{bmatrix}$$



Some special types of Matrices

Orthogonal Matrix

A square matrix is called orthogonal if $AA^T = I$

$$AA^T = I = A^T A$$

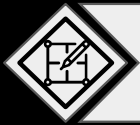
$$|AA^T| = |I|$$

$$|A||A^T| = 1$$

$$|A|^2 = 1$$

$$\Rightarrow |A| = \pm 1$$





Some special types of Matrices

Orthogonal Matrix

A square matrix is called orthogonal if $AA^T = I$



Remark

If A is an orthogonal matrix then its determinant must be ± 1



Some special types of Matrices

Try to observe directly that following are orthogonal matrices.

1.
$$\begin{bmatrix} \frac{1}{2} & \frac{-\sqrt{3}}{2} & 0 \\ \frac{\sqrt{3}}{2} & \frac{1}{2} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

2.
$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

3.
$$\begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{-1}{\sqrt{2}} \\ \frac{-1}{\sqrt{2}} & \frac{-1}{\sqrt{2}} \end{bmatrix}$$

Colⁿ/Row^s are the unit vectors

Q

Let $A = \begin{bmatrix} 0 & 2q & r \\ p & q & -r \\ p & -q & r \end{bmatrix}$. If $AA^T = I_3$, then $|p|$ is

mentos
Zindagi

$$\sqrt{0^2 + p^2 + p^2} = 1$$

$$\sqrt{2}|p| = 1$$

$$|p| = \frac{1}{\sqrt{2}}$$

☒ A $\frac{1}{\sqrt{2}}$

☐ B $\frac{1}{\sqrt{5}}$

☐ C $\frac{1}{\sqrt{6}}$

☐ D $\frac{1}{\sqrt{3}}$



Let $A = \begin{bmatrix} 0 & 2q & r \\ p & q & -r \\ p & -q & r \end{bmatrix}$. If $AA^T = I_3$, then $|p|$ is

- ☒ A $\frac{1}{\sqrt{2}}$
- ☐ B $\frac{1}{\sqrt{5}}$
- ☐ C $\frac{1}{\sqrt{6}}$
- ☐ D $\frac{1}{\sqrt{3}}$



Let $A = \begin{bmatrix} 0 & 2q & r \\ p & q & -r \\ p & -q & r \end{bmatrix}$. If $AA^T = I_3$, then $|p|$ is

Solution:

A is orthogonal matrix

$$\Rightarrow 0^2 + p^2 + p^2 = 1$$

$$\Rightarrow |p| = \frac{1}{\sqrt{2}}$$



If $\left\{ \sum_{i=1}^3 a_i b_i = \sum_{i=1}^3 b_i c_i = \sum_{i=1}^3 a_i c_i = 0 \right\}$ and $\left\{ \sum_{i=1}^3 a_i^2 = \sum_{i=1}^3 b_i^2 = \sum_{i=1}^3 c_i^2 = 1 \right\}$

then value of $\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} =$

col's are
per

col's
are unit
vectors

(A) 0

(B) 1

(C) -1

~~(D) ± 1~~



If $\sum_{i=1}^3 a_i b_i = \sum_{i=1}^3 b_i c_i = \sum_{i=1}^3 a_i c_i = 0$ and $\sum_{i=1}^3 a_i^2 = \sum_{i=1}^3 b_i^2 = \sum_{i=1}^3 c_i^2 = 1$

then value of $\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} =$

(A) 0

(B) 1

(C) -1

(D) ± 1

Solution:

$$\text{Let } A^T = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$$

$$\text{Now } A^T A = \begin{bmatrix} a_1^2 + a_2^2 + a_3^2 & a_1 b_1 + a_2 b_2 + a_3 b_3 & a_1 c_1 + a_2 c_2 + a_3 c_3 \\ a_1 b_1 + a_2 b_2 + a_3 b_3 & b_1^2 + b_2^2 + b_3^2 & b_1 c_1 + b_2 c_2 + b_3 c_3 \\ a_1 c_1 + a_2 c_2 + a_3 c_3 & b_1 c_1 + b_2 c_2 + b_3 c_3 & c_1^2 + c_2^2 + c_3^2 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I$$

Thus A is orthogonal matrix

Thus $|A| = \pm 1$

Q

Let $P = \begin{bmatrix} \frac{\sqrt{3}}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix}$, $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ and $Q = PAP^T$.

If $P^T Q^{2007} P = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, then $2a + b - 3c - 4d$ is equal to

Sol

$$P P^T = I = P^T P$$

$$\begin{aligned} P^T Q^{2007} P &= P^T (P A P^T)^{2007} P \\ &= P^T P A^{2007} P^T P \\ &= A^{2007} = \checkmark \end{aligned}$$

(A) 2004

(B) 2006

(C) 2007

(D) 2005

JEE Main 8th Apr, 2023



Let $P = \begin{bmatrix} \frac{\sqrt{3}}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix}$, $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ and $Q = PAP^T$.

If $P^T Q^{2007} P = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, then $2a + b - 3c - 4d$ is equal to

(A) 2004

(B) 2006

(C) 2007

(D) 2005



Let $P = \begin{bmatrix} \frac{\sqrt{3}}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix}$, $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ and $Q = PAP^T$.

If $P^T Q^{2007} P = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, then $2a + b - 3c - 4d$ is equal to

Solution:

$$\text{Here, } P = \begin{bmatrix} \frac{\sqrt{3}}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix}, A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

$$\text{Here, } PP^T = \begin{bmatrix} \frac{\sqrt{3}}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix} \begin{bmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$$

Solution:

Similarly $P^T P = 1$

$$\therefore Q = PAP^T$$

Now, $Q^{2007} = (PAP^T) (PAP^T) \dots 2007 \text{ times} = PA^{2007}P^T$

$$\text{As } A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

$$\Rightarrow A^2 = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1+0 & 1+1 \\ 0+0 & 0+1 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$$

$$A^3 = \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}$$

$$A^{2007} = \begin{bmatrix} 1 & 2007 \\ 0 & 1 \end{bmatrix}$$

Solution:

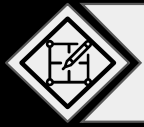
$$\text{Hence, } P^T Q^{2007} P = A^{2007} = \begin{bmatrix} 1 & 2007 \\ 0 & 1 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} 1 & 2007 \\ 0 & 1 \end{bmatrix}$$

$$\Rightarrow a = 1, b = 2007, c = 0, d = 1$$

$$\therefore 2a + b - 3c - 4d = 2(1) + 2007 - 3(0) - 4(1)$$

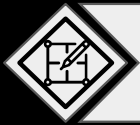
$$= 2 + 2007 - 4 = 2005$$



Some special types of Matrices

Unitary Matrix

A square matrix is called Unitary matrix if $A \times (\bar{A})^T = I$



Some special types of Matrices

Idempotent Matrix

A square matrix is called Idempotent matrix if $A^2 = A$

Clearly, A^n will also be equal to $A \forall n \geq 2$ |

$$A^2 = A$$

$$A^3 = A^2 = A$$

$$A^4 = A^2 = A$$



If $AB = A$ and $BA = B$ then show that B will be idempotent.

Already
done



If $AB = A$ and $BA = B$ then show that B will be idempotent.

Solution:

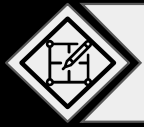
Given, $AB = A$

$$B(AB) = BA$$

$$B.B = B$$

$$B^2 = B$$

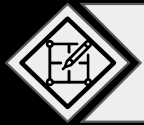
So B is idempotent



Some special types of Matrices

Involutory Matrix

A square matrix is called involutory if $A^2 = I$.



Some special types of Matrices

Nilpotent Matrix

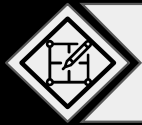
A square matrix is called nilpotent matrix of order m if:

$$A_{n \times n}^m = O_{n \times n} \text{ and } A_{n \times n}^{m-1} \neq O_{n \times n}$$



NOTE

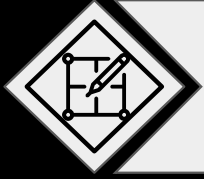
There is no way to check whether matrix is nilpotent or not, other than checking the powers manually.



Some special types of Matrices

Singular and Non-singular Matrix

A matrix is called singular if its determinant is zero, otherwise it is called non-singular.



Adjoint of Matrix

most
important



Adjoint of Matrix

For any square matrix, its adjoint is defined as transpose of its cofactor matrix.



Find the adjoint of :

(a) $A = \begin{bmatrix} 4 & 1 \\ -1 & 5 \end{bmatrix}$

(b) $B = \begin{bmatrix} 1 & -2 & 4 \\ 0 & 2 & 1 \\ -4 & 5 & 3 \end{bmatrix}$



Find the adjoint of:

$$(a) A = \begin{bmatrix} 4 & 1 \\ -1 & 5 \end{bmatrix}$$

$$\left[\begin{array}{cc} \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} & \begin{bmatrix} c_{11} & c_{12} \\ c_{21} & c_{22} \end{bmatrix} \\ \begin{bmatrix} \underline{a_{11}} & \underline{a_{12}} \\ a_{21} & a_{22} \end{bmatrix} & \begin{bmatrix} c_{11} & c_{21} \\ c_{12} & c_{22} \end{bmatrix} \end{array} \right]^T$$
$$\Rightarrow \begin{bmatrix} |A| & 0 \\ 0 & |A| \end{bmatrix}$$

$$\text{adj } A = \begin{bmatrix} 5 & -(-1) \\ -1 & 4 \end{bmatrix}^T$$

$$\text{coadj } A = \begin{bmatrix} 5 & -1 \\ 1 & 4 \end{bmatrix}$$

$$\begin{aligned} & \text{obv}^n A \times \text{adj } A \\ & \begin{bmatrix} 4 & 1 \\ -1 & 5 \end{bmatrix} \times \begin{bmatrix} 5 & -1 \\ 1 & 4 \end{bmatrix} \\ & = \begin{bmatrix} 21 & 0 \\ 0 & 21 \end{bmatrix} \\ & = 21 \times \hat{I} \\ & \quad \quad \quad \hookrightarrow |A| \end{aligned}$$



Find the adjoint of :

$$(a) A = \begin{bmatrix} 4 & 1 \\ -1 & 5 \end{bmatrix}$$

Solution:

The cofactors C_{ij} of A are:

$$C_{11} = 5, C_{12} = 1$$

$$C_{21} = -1, C_{22} = 4$$

$$\text{adj } A = \begin{bmatrix} 5 & 1 \\ -1 & 4 \end{bmatrix}^T = \begin{bmatrix} 5 & -1 \\ 1 & 4 \end{bmatrix}.$$

$$\begin{bmatrix} + & - & + \\ - & + & - \\ + & - & + \end{bmatrix}$$



Find the adjoint of:

$$(b) B = \begin{bmatrix} 1 & -2 & 4 \\ 0 & 2 & 1 \\ -4 & 5 & 3 \end{bmatrix}$$

$$\text{Cofactor matrix of } B = \begin{bmatrix} 1 & -4 & 8 \\ 26 & 15 & 3 \\ -10 & -1 & 2 \end{bmatrix}$$

$$\text{Adj } B = \begin{bmatrix} 1 & 26 & -10 \\ -4 & 15 & -1 \\ 8 & 3 & 2 \end{bmatrix}$$



Find the adjoint of :

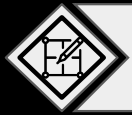
$$(b) \ B = \begin{bmatrix} 1 & -2 & 4 \\ 0 & 2 & 1 \\ -4 & 5 & 3 \end{bmatrix}$$

Solution:

$$B = \begin{bmatrix} 1 & -2 & 4 \\ 0 & 2 & 1 \\ -4 & 5 & 3 \end{bmatrix}$$

Its cofactor matrix is given by $\begin{bmatrix} 1 & -4 & 8 \\ 26 & 19 & 3 \\ -10 & -1 & 2 \end{bmatrix}$

$$\therefore \text{adj}B = \begin{bmatrix} 1 & -4 & 8 \\ 26 & 19 & 3 \\ -10 & -1 & 2 \end{bmatrix}' = \begin{bmatrix} 1 & 26 & -10 \\ -4 & 19 & -1 \\ 8 & 3 & 2 \end{bmatrix}$$



Adjoint of Matrix



Result

For any square matrix $A_{n \times n}$:

$$A \times (\text{adj } A) = |A| I_n = (\text{adj } A) \times A$$



Adjoint of Matrix

Properties of Adjoint



Adjoint of Matrix

Properties of Adjoint

For square matrix A & B of order n, we have:

1. $|\text{adj } A| = |A|^{n-1}$
 2. $\text{adj}(\text{adj } A) = |A|^{n-2} A$
 3. $|\text{adj}(\text{adj } A)| = |A|^{(n-1)^2}$
- $|\text{adj}(\text{adj}(\text{adj } A))| = |A|^{(n-1)^3}$



Adjoint of Matrix

Properties of Adjoint

4. $\text{adj}(A^T) = (\text{adj } A)^T$

5. $\text{adj}(KA) = K^{n-1} \text{adj } A$; K is constant

6. $\text{adj}(A^m) = (\text{adj } A)^m$; $m \in \mathbb{N}$

7. $\text{adj}(AB) = (\text{adj } B) (\text{adj } A)$

i.e. reversal law holds

$$|5A_3| = 5^3 |A_3|$$



If $|A| = 2$ where A is a square matrix of order 3 then find:

(a) $|\text{adj } A|$

(b) $|\text{adj}(\text{adj } A)|$

(c) $|\text{adj}(A^T)|$

(d) $|\text{adj}(5A)|$

$$(a) |\text{adj } A| = |A|^{3-1} = 2^2 = 4$$

$$(c) |(\text{adj } A)^T| = |\text{adj } A| = |A|^{3-1} = 2^2 = 4$$

$$(b) |\text{adj}(\text{adj } A)| = |A|^{(3-1)^2} = 2^4 = 16$$

$$(d) |\text{adj}(5A)| = |5A|^{3-1} = |5A|^2 = 5^6 \times |A|^2 = 5^6 \times 2^2$$



If $|A| = 2$ where A is a square matrix of order 3 then find:

(a) $|\text{adj } A|$

(b) $|\text{adj}(\text{adj } A)|$

(c) $|\text{adj}(A^T)|$

(d) $|\text{adj}(5A)|$

Solution:

(a) Given, $|A| = 2$

$$|\text{adj } A| = |A|^{n-1}$$

$$= (2)^{3-1}$$

$$= 2^2$$

$$= 4$$

(b) $|\text{adj}(\text{adj } A)|$

$$= |A|^{(n-1)^2}$$

$$= (2)^{(3-1)^2}$$

$$= 2^4$$

$$= 16$$



If $|A| = 2$ where A is a square matrix of order 3 then find:

(a) $|adj A|$

(b) $|adj (adj A)|$

(c) $|adj (A^T)|$

(d) $|adj (5A)|$

Solution:

$$\begin{aligned}(c) \quad |adj (A^T)| &= |adj(A)^T| \\ &= |adj(A)|^T \\ &= |adj(A)| \\ &= |A|^{n-1} \\ &= 2^{3-1} \\ &= 4\end{aligned}$$

$$\begin{aligned}(d) \quad |adj (5A)| &= |5^{n-1} adj A| \\ &= 5^{(3-1)^3} \times 4 \\ &= 5^6 \times 4\end{aligned}$$

Q

If A is a 3×3 matrix and $|A| = 2$, then $|3 \operatorname{adj}(|3A|A^2)|$ is equal to

$$= |3 \operatorname{adj}(|3A|A^2)|$$

$$= 3^3 |\operatorname{adj}(54A^2)|$$

$$= 3^3 \times |54A^2|^2$$

$$= 3^3 \times (54^3) |A^2|^2$$

$$= 3^3 \times 54^6 \times 2^4$$

$$|3A| = 3^3 |A| = 54 \quad \text{(A)} \quad 3^{12} \cdot 6^{10}$$

$$\text{(B)} \quad 3^{11} \cdot 6^{10}$$

$$\text{(C)} \quad 3^{12} \cdot 6^{11}$$

$$\text{(D)} \quad 3^{10} \cdot 6^{11}$$

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If A is a 3×3 matrix and $|A| = 2$, then $|3adj(|3A|A^2)|$ is equal to

(A) $3^{12}.6^{10}$

(B) $3^{11}.6^{10}$

(C) $3^{12}.6^{11}$

(D) $3^{10}.6^{11}$



If A is a 3×3 matrix and $|A| = 2$, then $|3adj(|3A|A^2)|$ is equal to

Solution:

$$\begin{aligned} |3adj(|3A|A^2)| &= 3^3 |adj(54A^2)| = 3^3 \cdot |54A^2|^2 \\ &= 3^3 \times 54^6 \times |A|^4 = 3^{11} \times 6^{10} \end{aligned}$$

Q

$$\textcircled{*} (\text{adj } kA_n) = k^{n-1} (\text{adj } A_n)$$

$$\textcircled{*} |k \times A_n| = k^n |A_n|$$

$$|2 \text{adj}(2 \text{adj}(2A))| = (32)^n$$

$$|2^3 \text{adj}(\text{adj}(2A))| = (32)^n$$

$$8^3 |\text{adj}(\text{adj}(2A))| = (32)^n$$

$$8^3 \times |2A|^4 = (32)^n \Rightarrow 8^3 \times (2^3)^4 \times 2^4 = (32)^n$$

Let $A = \begin{bmatrix} 1 & 2 & 3 \\ \alpha & 3 & 1 \\ 1 & 1 & 2 \end{bmatrix}$, $|A| = 2$. If $|2 \text{adj}(2 \text{adj}(2A))| = 32^n$, then $3n + \alpha$ is equal to

$$\begin{aligned} |A| &= 2 \\ \alpha &= -4 \end{aligned}$$

(A) 11

(B) 9

(C) 12

(D) 10



Let $A = \begin{bmatrix} 1 & 2 & 3 \\ \alpha & 3 & 1 \\ 1 & 1 & 2 \end{bmatrix}$, $|A| = 2$. If $|2adj(2adj(2A))| = 32^n$,
then $3n + \alpha$ is equal to

A 11

B 9

C 12

D 10

Solution:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ \alpha & 3 & 1 \\ 1 & 1 & 2 \end{bmatrix}$$

$$|A| = 2$$

$$\Rightarrow 1(6 - 1) - 2(2\alpha - 1) + 3(\alpha - 3) = 2$$

$$\Rightarrow 5 - 4\alpha + 2 + 3\alpha - 9 = 2$$

$$\Rightarrow -\alpha - 4 = 0$$

$$\Rightarrow \alpha = -4$$

$$\text{Now, } |2 \operatorname{adj}(2 \operatorname{adj}(2A))| = 32^n$$

$$\Rightarrow 2^3 |\operatorname{adj}(2 \operatorname{adj}(2A))| = 32^n$$

$$\Rightarrow 8 |\operatorname{Adj}(2 \operatorname{Adj}(2A))| = 32^n$$

$$\Rightarrow 8 |\operatorname{Adj}(2 \times 2^2 \operatorname{Adj}(A))| = 32^n$$

$$\Rightarrow 8 |\operatorname{Adj}(2^3 \operatorname{Adj}A)| = 32^n$$

$$\Rightarrow 8 |2^6 \operatorname{Adj}(\operatorname{Adj}A)| = 32^n$$

$$\Rightarrow 2^3 (2^6)^3 |\operatorname{Adj}(\operatorname{Adj})| = 32^n$$

$$\Rightarrow 2^3 \cdot 2^{18} |A|^4 = 32^n$$

$$\Rightarrow 2^{21} \cdot 2^4 = 32^n$$

$$\Rightarrow 2^{25} = 32^n$$

$$\Rightarrow (2^5)^5 = 32^n$$

$$\Rightarrow (32)^5 = 32^n$$

$$\therefore n = 5$$

$$\Rightarrow 3n + \alpha = 15 - 4 = 11$$



Let $A = \begin{pmatrix} 2 & -1 \\ 0 & 2 \end{pmatrix}$.

If $B = I - {}^5C_1(\text{adj } A) + {}^5C_2(\text{adj } A)^2 - \dots - {}^5C_5(\text{adj } A)^5$,
then the sum of all elements of the matrix B is:

$$\begin{aligned} B &= (I - \text{adj } A)^5 \\ &= \left(\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix} \right)^5 \\ &= \begin{bmatrix} -1 & -1 \\ 0 & -1 \end{bmatrix}^5 \end{aligned}$$

(A) -5

(B) -6

(C) -7

(D) -8



Let $A = \begin{pmatrix} 2 & -1 \\ 0 & 2 \end{pmatrix}$.

If $B = I - {}^5C_1(\text{adj } A) + {}^5C_2(\text{adj } A)^2 - \dots - {}^5C_5(\text{adj } A)^5$,
then the sum of all elements of the matrix B is:

(A) -5

(B) -6

(C) -7

(D) -8



Let $A = \begin{pmatrix} 2 & -1 \\ 0 & 2 \end{pmatrix}$.

If $B = I - {}^5C_1(\text{adj } A) + {}^5C_2(\text{adj } A)^2 - \dots - {}^5C_5(\text{adj } A)^5$,
then the sum of all elements of the matrix B is:

Solution:

$$B = (I - \text{adj } A)^5 = \begin{bmatrix} -1 & -1 \\ 0 & -1 \end{bmatrix}^5 = \begin{bmatrix} -1 & -5 \\ 0 & -1 \end{bmatrix}$$

Sum of its all elements = -7 .



If A is $n \times n$ matrix & B is its adjoint then prove that:

$$|AB + KI_n| = (|A| + K)^n; K : \text{constant}$$

$$\begin{aligned} \text{Ans } |AB + KI_n| &= |A \times \text{adj} A + KI_n| \\ &= |\underbrace{|A|}_{\text{scalar}} \underbrace{I_n}_{\text{matrix}} + \underbrace{K}_{\text{scalar}} \underbrace{I_n}_{\text{matrix}}| \\ &= |(\underbrace{|A| + K}_{\text{scalar}}) I_n| \\ &= (|A| + K)^n \end{aligned}$$

$$\begin{aligned} |5I_n| &= 5^n \\ \text{or } 5^n \times |I_n| \end{aligned}$$

Recall

For any square matrix $A_{n \times n}$:

$$A \times (\text{adj } A) = |A| I_n = (\text{adj } A) \times A$$



If A is $n \times n$ matrix & B is its adjoint then prove that:

$$|AB + KI_n| = (|A| + K)^n; K : \text{constant}$$

Solution:

$$\text{So, } AB = |A| \cdot I_n$$

$$|AB + KI_n| = ||A|I_n + KI_n|$$

$$= |(|A| + K)I_n|$$

$$= (|A| + K)^n |I_n|$$

$$= (|A| + K)^n$$



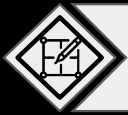
Adjoint of Matrix

Let's prove few results, for better understanding :

For square matrix A & B of order n, we have:

$$1. \underline{|\text{adj } A| = |A|^{n-1}}$$

$$\begin{aligned} \text{Pf } A_n \times \text{adj } A_n &= |A_n| \times I_n \\ |A| \times |\text{adj } A| &= |\underline{|A|} \times I_n| \\ |A| \times |\text{adj } A| &= |A|^n \times |I_n| \\ |\text{adj } A| &= |A|^{n-1} \end{aligned}$$



Adjoint of Matrix

Let's prove few results, for better understanding :

For square matrix A & B of order n, we have:

$$2. \text{adj}(\text{adj } A) = |A|^{n-2} A$$

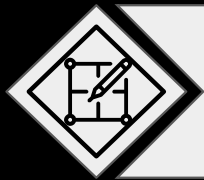
get rid of it.

$$\text{P1} \quad (\text{adj } A) \times \text{adj}(\text{adj } A) = |A|^{n-1} I_n$$

Note

$$A(\text{adj } A) \times \text{adj}(\text{adj } A) = |A|^{n-1} \times A$$
$$|A| \times I \times \text{adj}(\text{adj } A) = |A|^{\frac{n-2}{A}} \times A$$

~~_____~~



Inverse of Matrix


$$AB = BA = I$$

Formula for A^{-1}

$$A \times \text{adj } A = |A| \times \overline{I}$$
$$A \times \left(\frac{1}{|A|} \text{adj } A \right) = \overline{I}$$

← A^{-1}

Clearly, A^{-1} exists (i.e. matrix A is invertible iff A is non-singular)



Find the inverse of:

$$(a) A = \begin{bmatrix} 2 & 5 \\ -3 & 1 \end{bmatrix}$$

$$(b) B = \begin{bmatrix} 1 & -1 & 2 \\ 0 & 2 & -3 \\ 3 & -2 & 4 \end{bmatrix}$$



Find the inverse of:

$$(a) A = \begin{bmatrix} 2 & 5 \\ -3 & 1 \end{bmatrix}$$

$$A^{-1} = \frac{1}{17} \begin{bmatrix} 1 & -5 \\ 3 & 2 \end{bmatrix}$$



Find the inverse of:

$$(a) A = \begin{bmatrix} 2 & 5 \\ -3 & 1 \end{bmatrix}$$

Solution:

$$\text{Let } A = \begin{bmatrix} 2 & 5 \\ -3 & 1 \end{bmatrix}.$$

$$\therefore |A| = \begin{vmatrix} 2 & 5 \\ -3 & 1 \end{vmatrix} = 2 + 15 = 17$$

$\therefore A^{-1}$ exists.

$$\text{adj } A = \begin{bmatrix} 1 & -5 \\ 3 & 2 \end{bmatrix}$$

$$\text{Hence } A^{-1} = \frac{1}{|A|} \cdot \text{adj } A = \frac{1}{17} \begin{bmatrix} 1 & -5 \\ 3 & 2 \end{bmatrix}$$



Find the inverse of:

$$(b) \ B = \begin{bmatrix} 1 & -1 & 2 \\ 0 & 2 & -3 \\ 3 & -2 & 4 \end{bmatrix}$$

HW



Find the inverse of:

$$(b) \ B = \begin{bmatrix} 1 & -1 & 2 \\ 0 & 2 & -3 \\ 3 & -2 & 4 \end{bmatrix}$$

Solution:

$$|B| = \begin{vmatrix} 1 & -1 & 2 \\ 0 & 2 & -3 \\ 3 & -2 & 4 \end{vmatrix} = 2 + 9 - 12 = -1 \neq 0$$

$\therefore B^{-1}$ exists

$$\text{Now } Adj\ B = \begin{bmatrix} 2 & -9 & -6 \\ 0 & -2 & -1 \\ -1 & 3 & 2 \end{bmatrix}^T = \begin{bmatrix} 2 & 0 & -1 \\ -9 & -2 & 3 \\ -6 & -1 & 2 \end{bmatrix}$$

$$\text{Hence } A^{-1} = \frac{1}{|A|} \cdot adj\ A = - \begin{bmatrix} 2 & 0 & -1 \\ -9 & -2 & 3 \\ -6 & -1 & 2 \end{bmatrix} = \begin{bmatrix} -2 & 0 & 1 \\ 9 & 2 & -3 \\ 6 & 1 & -2 \end{bmatrix}$$



$A = \begin{bmatrix} 1 & 0 & -K \\ 2 & 1 & 2 \\ K & 0 & 1 \end{bmatrix}$ is invertible for

- ☐ A $K = 1$
- ☐ B $K = 0$
- ☐ C $K = -1$
- ☐ D $K \in \mathbb{R}$



$A = \begin{bmatrix} 1 & 0 & -K \\ 2 & 1 & 2 \\ K & 0 & 1 \end{bmatrix}$ is invertible for

(A) $K = 1$

(B) $K = 0$

(C) $K = -1$

✓ (D) $K \in \mathbb{R}$



$$A = \begin{bmatrix} 1 & 0 & -K \\ 2 & 1 & 2 \\ K & 0 & 1 \end{bmatrix} \text{ is invertible for}$$

Solution:

A is invertible, if

$$|A| \neq 0$$

$$1(1 - 0) - 0(2 - 3K) - K(0 - K) \neq 0$$

$$1 + K^2 \neq 0$$

$$K \in \mathbb{R}$$

Thus, it is non-zero for all values

Multiple correct question

JEE Advanced 2014, P1



Let M be a 2×2 symmetric matrix with integer entries.
Then, M is invertible, if

Sol $M = \begin{bmatrix} a & b \\ b & c \end{bmatrix}$

$ac - b^2 \neq 0$

(c) $b = 0$
 $a, c \neq 0$ $\rightarrow \checkmark$

(d) $ac = \text{not a perfect sq.}$ $\rightarrow \checkmark$



The first column of M is the transpose of the second row of M $a = b$



The second row of M is the transpose of the first column of M $b = c$
 $a = b = c$



M is the diagonal matrix with non-zero entries in the main diagonal



The product of entries in the main diagonal of M is not the square of an integer

Multiple correct question

JEE Advanced 2014, P1



Let M be a 2×2 symmetric matrix with integer entries.
Then, M is invertible, if

- ☐ A The first column of M is the transpose of the second row of M
- ☐ B The second row of M is the transpose of the first column of M
- ☒ C M is the diagonal matrix with non-zero entries in the main diagonal
- ☒ D The product of entries in the main diagonal of M is not the square of an integer

Multiple correct question

JEE Advanced 2014, P1



Let M be a 2×2 symmetric matrix with integer entries.
Then, M is invertible, if

Solution:

$$\text{Let } M = \begin{bmatrix} a & b \\ b & c \end{bmatrix}$$

$$(a) \text{ Given that } \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} b \\ c \end{bmatrix}$$

$$\Rightarrow a = b = c = \alpha \text{ (let)}$$

$$\Rightarrow M = \begin{bmatrix} \alpha & \alpha \\ \alpha & \alpha \end{bmatrix}$$

$$\Rightarrow |M| = 0$$

$$\Rightarrow M \text{ is non-invertible.}$$

Solution:

(b) Given that $[bc] = [ab]$

$$\Rightarrow a = b = c = \alpha \text{ (let)}$$

$$\text{Again } |M| = 0$$

$\Rightarrow M$ is non-invertible.

$$(c) \text{ As given } M = \begin{bmatrix} a & 0 \\ 0 & c \end{bmatrix}$$

$$\Rightarrow |M| = ac \neq 0 \text{ (}\because a \text{ and } c \text{ are non-zero)}$$

$\Rightarrow M$ is invertible.

$$(d) M = \begin{bmatrix} a & b \\ b & c \end{bmatrix} \Rightarrow |M| = ac - b^2 \neq 0$$

$\because ac$ is not equal to square of an integer.

$\therefore M$ is invertible.

Q

If A & B are non-zero matrices, such that $AB = O$, then

Sol. $AB = 0$

$$|A||B| = 0$$

$$|A\rangle = 0 \text{ or } |B\rangle = 0$$

Say, $|A| = 0$ & $|B| \neq 0$

$\hookrightarrow B$ invertible

Here, $AB = 0$

$$AB B^{-1} = O \times B^{-1}$$

$A=0$ & contradⁿ.

A A or B is singular

 A and B are singular

Ⓒ Only A is singular

D Only B is singular

$$\Rightarrow |A|=0 \text{ \& } |B|=0$$



If A & B are non-zero matrices, such that $AB = O$, then

- ☐ A A or B is singular
- ☒ B A and B are singular
- ☐ C Only A is singular
- ☐ D Only B is singular



Inverse of Matrix

Properties of Inverse

$$1. (A^{-1})^{-1} = A$$

$$2. |A^{-1}| = \frac{1}{|A|}$$

$$3. (A^T)^{-1} = (A^{-1})^T \quad ; \quad A^{-n} = (A^{-1})^n$$

$$4. (AB)^{-1} = B^{-1}A^{-1} \quad \text{i.e. reversal law holds.}$$

$$A A^{-1} = I$$

$$|A| |A^{-1}| = |I|$$

$$|A^{-1}| = \frac{1}{|A|}$$



If A is a 3×3 non-singular matrix such that $AA^T = A^T A$ and $B = A^{-1}A^T$ then $BB^T =$

$$\begin{aligned} BB^T &= (A^{-1}A^T)(A^{-1}A^T)^T \\ &= A^{-1}A^T \times (A^T)^T(A^{-1})^T \\ &= A^{-1}\underline{A^T \times A}(A^{-1})^T \\ &= A^{-1}A \times A^T(A^T)^{-1} \\ &= I \end{aligned}$$

(A) $I + B$

~~(B) I~~

(C) B^{-1}

(D) $(B^{-1})^T$



If A is a 3×3 non-singular matrix such that $AA^T = A^T A$ and $B = A^{-1}A^T$ then $BB^T =$

☐ A $I + B$

☒ B I

☐ C B^{-1}

☐ D $(B^{-1})^T$



If A is a 3×3 non-singular matrix such that $AA^T = A^T A$ and $B = A^{-1}A^T$ then $BB^T =$

Solution:

Taking transpose on both sides

$$B^T = (A^{-1} A^T)^T$$

$$B^T = (A^T)^T (A^{-1})^T = A(A^{-1})^T$$

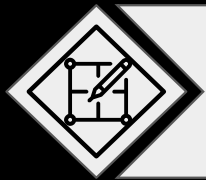
$$\text{Now, } B \cdot B^T = (A^{-1} A^T) (A \cdot (A^{-1})^T)$$

$$= A^{-1}(A^T A) (A^{-1})^T$$

$$= A^{-1}(AA^T) (A^{-1})^T$$

$$= (A^{-1}A) (A^T (A^{-1})^T)$$

$$= I \cdot I = I$$



Elementary Transformations

$$\begin{bmatrix} 8 & 10 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 2 & 3 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 2 & 4 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -1 \\ 8 & 10 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 2 & 4 \end{bmatrix}$$

$$\begin{bmatrix} 9 & 9 \\ 8 & 10 \end{bmatrix} = \begin{bmatrix} 3 & 3 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 2 & 4 \end{bmatrix}$$



Elementary Transformations

Following three transformations are called elementary row (column) transformations.

1. $R_i \leftrightarrow R_j, (C_i \leftrightarrow C_j)$

2. $(R_i \rightarrow KR_i) (C_i \rightarrow KC_i) ; K: \text{Non zero constant}$

3. $R_i \rightarrow R_i + KR_j, (C_i \rightarrow C_i + KC_j) ; K: \text{Non zero constant}$

Two matrices are said to be equivalent if one is obtained from other using elementary transformation.



Elementary Transformations

Theorem: Let $C = A \times B$ be a product of two matrices. Any elementary row (column) transformation of C can be obtained by applying same elementary row (column) transformation on pre-factor (post factor).



Elementary Transformations

Theorem: Let $C = A \times B$ be a product of two matrices. Any elementary row (column) transformation of C can be obtained by applying same elementary row (column) transformation on pre-factor (post factor).

Elementary Matrix: The matrix obtained by applying a single elementary transformation on identity matrix is called elementary matrix.



Elementary Transformations

Finding Inverse using Elementary transformations

Let A be the matrix whose inverse is to be formed.



Elementary Transformations

Finding Inverse using Elementary transformations

Let A be the matrix whose inverse is to be formed.

Algorithm:

Step 1: Write $A = IA$

Step 2: Perform elementary row transformations successively on A in LHS & same transformation on prefactor I on RHS till we obtain $I = BA$.

Step 3: B is the required A^{-1}

For finding A^{-1} using elementary column transformations, write $A = AI$.



Find the inverse of:

(a) $A = \begin{bmatrix} 2 & 1 \\ 3 & 4 \end{bmatrix}$

(b) $\begin{bmatrix} 1 & 2 & -2 \\ -1 & 3 & 0 \\ 0 & -2 & 1 \end{bmatrix}$

a_{ii} ko 1 banao, reciprocal se multiply krke, fir aage zeros banao



Find the inverse of:

$$(a) A = \begin{bmatrix} 2 & 1 \\ 3 & 4 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 1 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 3 & 4 \end{bmatrix}$$

$$R_1 \rightarrow \frac{1}{2} R_1$$

$$\begin{bmatrix} 1 & \frac{1}{2} \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & 0 \\ 0 & 1 \end{bmatrix} \times A$$

$$R_2 \rightarrow R_2 - 3R_1$$

$$\begin{bmatrix} 1 & \frac{1}{2} \\ 0 & \frac{5}{2} \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & 0 \\ -\frac{3}{2} & 1 \end{bmatrix} A$$

$$\begin{bmatrix} 1 & \frac{1}{2} \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & 0 \\ -\frac{3}{2} & 1 \end{bmatrix} \times A$$
$$R_1 \rightarrow R_1 - \frac{1}{2} R_2$$

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} - & - \\ - & - \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 3 & 4 \end{bmatrix}$$

Ans

Solution:

Taking $A = IA$

$$\begin{bmatrix} 2 & 1 \\ 3 & 4 \end{bmatrix} = A \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad \left(R_1 \rightarrow \frac{1}{2}R_1 \right)$$

$$\begin{bmatrix} 1 & \frac{1}{2} \\ 0 & \frac{-5}{2} \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & 0 \\ 0 & 1 \end{bmatrix} A \quad (R_2 \rightarrow 3R_1 - R_2)$$

$$\begin{bmatrix} 1 & \frac{1}{2} \\ 0 & \frac{-5}{2} \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & 0 \\ \frac{3}{2} & -1 \end{bmatrix} A \quad \left(R_2 \rightarrow -\frac{2}{5}R_2 \right)$$

$$\begin{bmatrix} 1 & \frac{1}{2} \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & 0 \\ \frac{-3}{5} & \frac{2}{5} \end{bmatrix} A \quad \left(R_1 \rightarrow R_1 - \frac{1}{2}R_2 \right)$$

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} \frac{4}{5} & \frac{-1}{5} \\ \frac{-3}{5} & \frac{2}{5} \end{bmatrix} A$$

$$\text{Thus } A^{-1} = \begin{bmatrix} \frac{4}{5} & \frac{-1}{5} \\ \frac{-3}{5} & \frac{2}{5} \end{bmatrix}$$



Find the inverse of:

(b)
$$\begin{bmatrix} 1 & 2 & -2 \\ -1 & 3 & 0 \\ 0 & -2 & 1 \end{bmatrix}$$

Solution:

$$\begin{bmatrix} 1 & 2 & -2 \\ -1 & 3 & 0 \\ 0 & -2 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} A \quad (R_2 \rightarrow R_1 + R_2)$$

$$\begin{bmatrix} 1 & 2 & -2 \\ 0 & 5 & -2 \\ 0 & -2 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} A \quad \left(R_2 \rightarrow \frac{1}{5}R_2\right)$$

$$\begin{bmatrix} 1 & 2 & -2 \\ 0 & 1 & -\frac{2}{5} \\ 0 & -2 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ \frac{1}{5} & \frac{1}{5} & 0 \\ 0 & 0 & 1 \end{bmatrix} A \quad (R_1 \rightarrow 2R_2 - R_1, R_3 \rightarrow 2R_2 + R_3)$$

$$\begin{bmatrix} 1 & 0 & \frac{6}{5} \\ 0 & 1 & -\frac{2}{5} \\ 0 & 0 & \frac{1}{5} \end{bmatrix} = \begin{bmatrix} \frac{-3}{5} & \frac{2}{5} & 0 \\ \frac{1}{5} & \frac{1}{5} & 0 \\ \frac{2}{5} & \frac{2}{5} & 1 \end{bmatrix} A \quad (R_3 \rightarrow 5R_3)$$

Solution:

$$\begin{bmatrix} 1 & 0 & \frac{6}{5} \\ 0 & 1 & \frac{-2}{5} \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} \frac{-3}{5} & \frac{2}{5} & 0 \\ \frac{1}{5} & \frac{1}{5} & 0 \\ 2 & 2 & \frac{1}{5} \end{bmatrix} A$$

$$R_1 \rightarrow \frac{6}{5}R_3 - R_1, R_2 \rightarrow \frac{2}{5}R_3 + R_2$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 3 & 2 & \frac{6}{25} \\ 1 & 1 & \frac{2}{25} \\ 2 & 2 & \frac{1}{5} \end{bmatrix} A$$

$$A^{-1} = \begin{bmatrix} 3 & 2 & \frac{6}{25} \\ 1 & 1 & \frac{2}{25} \\ 2 & 2 & \frac{1}{5} \end{bmatrix}$$

Q

Find A^{-1} using column transformation:

$$A = \begin{bmatrix} 0 & 1 \\ 3 & 4 \end{bmatrix}$$

Solution:

$$A = A \times I$$

$$\begin{bmatrix} 0 & 1 \\ 3 & 4 \end{bmatrix} = A \times \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

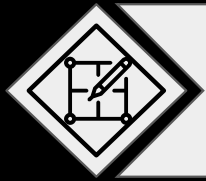
$$C_1 \leftrightarrow C_2$$

$$\begin{bmatrix} 1 & 0 \\ 4 & 3 \end{bmatrix} = A \times \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$C_1 \rightarrow \frac{1}{3}C_1$$

$$\begin{bmatrix} 1 & 0 \\ 4 & 1 \end{bmatrix} = A \times \begin{bmatrix} 0 & 1/3 \\ 1 & 0 \end{bmatrix}$$

$$C_1 \rightarrow C_1 - 4C_2$$



System of linear Equations using Matrices

$$\begin{aligned} AX &= B \\ X &= A^{-1}B \end{aligned}$$



Solve the following system of linear equations:

$$2x + 5y = 1; 3x + 2y = 7.$$

$$\begin{cases} 2x + 5y = 1 \\ 3x + 2y = 7 \end{cases}$$

$$\rightarrow \begin{bmatrix} 2 & 5 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 \\ 7 \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2 & 5 \\ 3 & 2 \end{bmatrix}^{-1} \begin{bmatrix} 1 \\ 7 \end{bmatrix}$$



Solve the following system of linear equations:
 $2x + 5y = 1$; $3x + 2y = 7$.

Solution:

Given, $2x + 5y = 1$; $3x + 2y = 7$

$$\begin{bmatrix} 2 & 5 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 \\ 7 \end{bmatrix} \quad (AX = B)$$

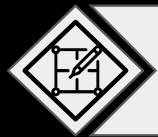
$$|A| = \begin{vmatrix} 2 & 5 \\ 3 & 2 \end{vmatrix} = -11 \neq 0$$

$$A^{-1} = -\frac{1}{11} \begin{bmatrix} 2 & -5 \\ -3 & 2 \end{bmatrix}$$

$$\Rightarrow X = A^{-1}B$$

$$\begin{bmatrix} x \\ y \end{bmatrix} = -\frac{1}{11} \begin{bmatrix} 2 & -5 \\ -3 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 7 \end{bmatrix} = -\frac{1}{11} \begin{bmatrix} -33 \\ 11 \end{bmatrix} = \begin{bmatrix} 3 \\ -1 \end{bmatrix}$$

$$\Rightarrow x = 3, y = -1$$



System of linear Equations using Matrices

Consider,

$$a_1x + b_1y + c_1z = d_1$$

$$a_2x + b_2y + c_2z = d_2$$

$$a_3x + b_3y + c_3z = d_3$$

Clearly, we can represent it as: $AX = B$

$$\text{Where, } A = \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, B = \begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix}$$

$$\text{So, } X = A^{-1}B$$

$$X = \frac{(\text{adj}A) \times B}{|A|}$$



Solve the following system of equations, using matrix method:

$$x + 2y + z = 7, x + 3z = 11, 2x - 3y = 1.$$

$$A X = B$$

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 1 & 0 & 3 \\ 2 & -3 & 0 \end{bmatrix}$$

$$X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$B = \begin{bmatrix} 7 \\ 11 \\ 1 \end{bmatrix}$$

$$X = A^{-1} B$$

$$X = \left(\frac{1}{|A|} \times \text{adj } A \right) \times B$$



Solve the following system of equations, using matrix method:

$$x + 2y + z = 7, x + 3z = 11, 2x - 3y = 1.$$

Solution:

The given system of equations is

$$x + 2y + z = 7$$

$$x + 0y + 3z = 11$$

$$2x - 3y + 0z = 1 \text{ or}$$

$$\begin{bmatrix} 1 & 2 & 1 \\ 1 & 0 & 3 \\ 2 & -3 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 7 \\ 11 \\ 1 \end{bmatrix} \text{ or } AX = B, \text{ where}$$

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 1 & 0 & 3 \\ 2 & -3 & 0 \end{bmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \text{ and } B = \begin{bmatrix} 7 \\ 11 \\ 1 \end{bmatrix}$$

$$\text{Now, } |A| = \begin{vmatrix} 1 & 2 & 1 \\ 1 & 0 & 3 \\ 2 & -3 & 0 \end{vmatrix} = 18$$

Solution:

So, the given system of equations has a unique solution given by $X = A^{-1}B$.

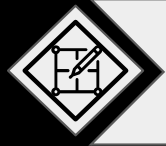
$$\text{adj}A = \begin{bmatrix} 9 & 6 & -3 \\ -3 & -2 & 7 \\ 6 & -2 & -2 \end{bmatrix}^T = \begin{bmatrix} 9 & -3 & 6 \\ 6 & -2 & -2 \\ -3 & 7 & -2 \end{bmatrix}$$

$$\Rightarrow A^{-1} = \frac{1}{|A|} \text{adj}A = \frac{1}{18} \begin{bmatrix} 9 & -3 & 6 \\ 6 & -2 & -2 \\ -3 & 7 & -2 \end{bmatrix}$$

Now, $X = A^{-1}B$

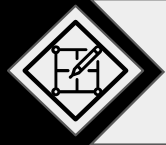
$$\Rightarrow X = \frac{1}{18} \begin{bmatrix} 9 & -3 & 6 \\ 6 & -2 & -2 \\ -3 & 7 & -2 \end{bmatrix} \begin{bmatrix} 7 \\ 11 \\ 1 \end{bmatrix} = \frac{1}{18} \begin{bmatrix} 63 - 33 + 6 \\ 42 - 22 - 2 \\ -21 + 77 - 2 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{18} \begin{bmatrix} 36 \\ 18 \\ 54 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix} \Rightarrow x = 2, y = 1, z = 3$$



System of linear Equations using Matrices

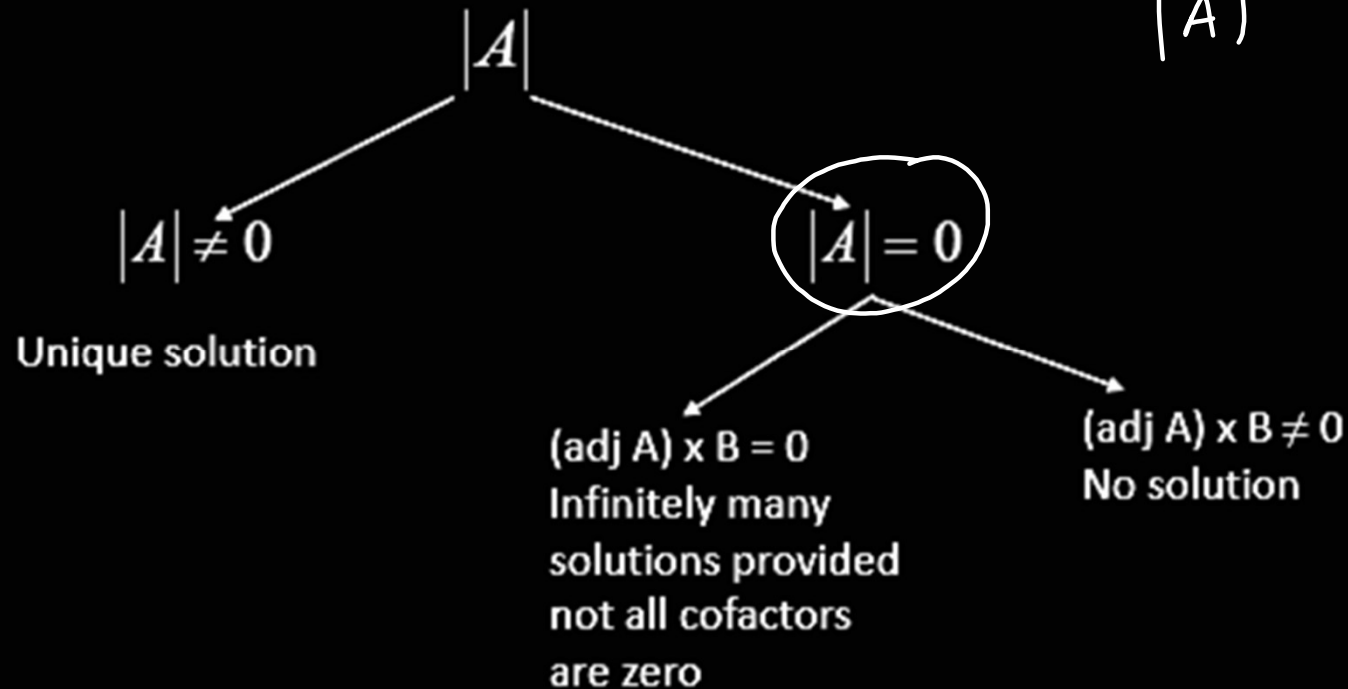
Number of solutions

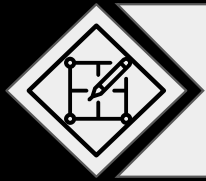


System of linear Equations using Matrices

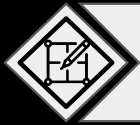
Number of solutions

$$X = \frac{1}{|A|} (\text{adj } A) \times B$$





Cayley Hamilton Theorem



Cayley Hamilton Theorem

Characteristic Equations

If A is any square matrix, then :

$$|A - xI| = 0$$

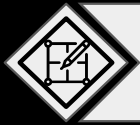
is called its characteristic equation.

Eg. characteristic equation of $\begin{bmatrix} 2 & 3 \\ 1 & 0 \end{bmatrix}$ is:

$$A - xI = \begin{bmatrix} 2 & 3 \\ 1 & 0 \end{bmatrix} - \begin{bmatrix} x & 0 \\ 0 & x \end{bmatrix}$$

$$A - xI = \begin{bmatrix} 2-x & 3 \\ 1 & -x \end{bmatrix}$$

n



Cayley Hamilton Theorem

Characteristic Equations

If A is any square matrix, then :

$$|A - xI| = 0$$

is called its characteristic equation.

Eg. characteristic equation of $\begin{bmatrix} 2 & 3 \\ 1 & 0 \end{bmatrix}$ is:

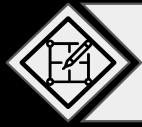
$$|A - xI| = 0$$

$$\Rightarrow \begin{vmatrix} 2-x & 3 \\ 1 & -x \end{vmatrix} = 0$$

$$\Rightarrow -2x + x^2 - 3 = 0$$

$$\text{i.e. } x^2 - 2x - 3 = 0 \quad \checkmark$$

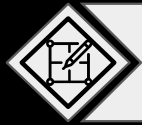
$$A^2 - 2A - 3I = 0$$



Cayley Hamilton Theorem

Cayley Hamilton Theorem:

Every Matrix satisfies its characteristic equation.



Cayley Hamilton Theorem

Cayley Hamilton Theorem:

Every Matrix satisfies its characteristic equation.

Characteristic equation of $\begin{bmatrix} 2 & 3 \\ 1 & 0 \end{bmatrix}$ is:

$$|A - xI| = 0$$

$$\Rightarrow \begin{vmatrix} 2-x & 3 \\ 1 & -x \end{vmatrix} = 0$$

$$\Rightarrow -2x + x^2 - 3 = 0$$

$$\text{i.e. } x^2 - 2x - 3 = 0$$

Using Cayley Hamilton then, we have

$$A^2 - 2A - 3I = 0.$$

Q

If $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & -2 & 4 \end{bmatrix}$ and $A^{-1} = \frac{1}{6}(A^2 + cA + dI)$, then find c and d .

mentos
Zindagi

$$6I = A^3 + cA^2 + dA$$

$$A^3 + cA^2 + dA - 6I = 0$$

Char eqn. of A

$$|A - \lambda I| = 0$$

$$\begin{vmatrix} 1-\lambda & 0 & 0 \\ 0 & 1-\lambda & 1 \\ 0 & -2 & 4-\lambda \end{vmatrix} = 0$$

$$\lambda^3 - 6\lambda^2 + 11\lambda - 6 = 0$$

CH thm. —

$$A^3 - 6A^2 + 11A - 6I = 0$$

$$c = -6 \quad \text{--- } |A|$$

$$d = 11$$



If $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & -2 & 4 \end{bmatrix}$ and $A^{-1} = \frac{1}{6}(A^2 + cA + dI)$, then find **c** and **d**.

Solution:

Its characteristic equation

$$|A - xI| = 0$$

$$\left| \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & -2 & 4 \end{bmatrix} - \begin{bmatrix} x & 0 & 0 \\ 0 & x & 0 \\ 0 & 0 & x \end{bmatrix} \right| = 0$$

$$\begin{vmatrix} 1-x & 0 & 0 \\ 0 & 1-x & 1 \\ 0 & -2 & 4-x \end{vmatrix} = 0$$

$$x^3 - 6x^2 + 11x - 6 = 0$$

Solution:

$$\text{Also, } A^3 - 6A^2 + 11A - 6I = 0$$

Multiply A^{-1} both sides

$$A^{-1}A^3 - 6A^{-1}A^2 + 11A^{-1}A - 6A^{-1}I = A^{-1}0$$

$$A^2 - 6A + 11I - 6A^{-1} = 0$$

$$6A^{-1} = A^2 - 6A + 11I$$

$$A^{-1} = \frac{1}{6} [A^2 - 6A + 11I]$$

Comparing with

$$A^{-1} = \frac{1}{6} (A^2 + cA + dI)$$

$$c = -6, d = 11$$



NOTE

- For a 2×2 matrix $A_{2 \times 2}$, characteristic equation is given by

$$\lambda^2 - [\text{Tr}(A)]\lambda + |A| = 0$$

- For a 3×3 matrix $A_{3 \times 3}$, characteristic equation is given by

$$\lambda^3 - [\text{Tr}(A)]\lambda^2 + \frac{1}{2}\lambda[(\text{Tr}(A))^2 - \text{Tr}(A^2)] - |A| = 0$$



Let $A = \begin{pmatrix} 1 & 2 \\ -2 & -5 \end{pmatrix}$. Let $\alpha, \beta \in \mathbb{R}$ be such that $\alpha A^2 + \beta A = 2I$. Then $\alpha + \beta$ is equal to

$$\alpha A^2 + \beta A - 2I = 0$$

$$A^2 + \frac{\beta}{\alpha} A - \frac{2}{\alpha} I = 0$$

$$-\text{Tr}(A) = \beta/\alpha \Rightarrow 4 = \beta/\alpha \Rightarrow \beta = 4\alpha$$

$$|A| = -2/\alpha \Rightarrow -1 = -2/\alpha$$

$\alpha = 2$

(A) -10

(B) -6

(C) 6

(D) 10

JEE Main 27th July, 2022



Let $A = \begin{pmatrix} 1 & 2 \\ -2 & -5 \end{pmatrix}$. Let $\alpha, \beta \in \mathbb{R}$ be such that $\alpha A^2 + \beta A = 2I$. Then $\alpha + \beta$ is equal to

☐ A -10

☐ B -6

☐ C 6

☒ D 10



Let $A = \begin{pmatrix} 1 & 2 \\ -2 & -5 \end{pmatrix}$. Let $\alpha, \beta \in \mathbb{R}$ be such that $\alpha A^2 + \beta A = 2I$. Then $\alpha + \beta$ is equal to

Solution:

Characteristic equation of matrix A is

$$|A - \lambda I| = 0$$

$$\begin{vmatrix} 1 - \lambda & 2 \\ -2 & -5 - \lambda \end{vmatrix} = 0$$

$$\Rightarrow \lambda^2 + 4\lambda = 1$$

$$\Rightarrow A^2 + 4A = I$$

$$\Rightarrow 2A^2 + 8A = 2I$$

On comparing with $\alpha A^2 + \beta A = 2I$, we get

$$\alpha = 2, \beta = 8$$

$$\therefore \alpha + \beta = 10$$



Let $M = \begin{bmatrix} \sin^4 \theta & -1 - \sin^2 \theta \\ 1 + \cos^2 \theta & \cos^4 \theta \end{bmatrix} = \alpha I + \beta M^{-1}$, where $\alpha = \alpha(\theta)$ & $\beta = \beta(\theta)$ are real numbers, and I is the 2×2 identity matrix. If α^* is the minimum of the set $\{\alpha(\theta) : \theta \in [0, 2\pi)\}$ and β^* is the minimum of the set $\{\beta(\theta) : \theta \in [0, 2)\}$, then the value of $\alpha^* + \beta^*$ is

$$M = \alpha I + \beta M^{-1}$$

$$M^2 = \alpha M + \beta I$$

$$M^2 - \alpha M - \beta I = 0$$

$$\theta_{\min} = \frac{1}{2} = \alpha^*$$

$$\alpha = \text{Tr}(M) = \sin^4 \theta + \cos^4 \theta$$

$$-\beta = |M| \Rightarrow \beta = - (2 + \sin^2 \theta \cos^2 \theta + \sin^4 \theta \cos^4 \theta)$$

For $\theta \in [0, \frac{\pi}{2}]$ $\textcircled{A} - \frac{37}{16}$

$\sin^4 \theta + \cos^4 \theta$ ke max aur min $\textcircled{B} - \frac{31}{16}$

hamesha $\theta = 0$ & $\theta = \frac{\pi}{2}$ $\textcircled{C} - \frac{17}{16}$

$\textcircled{D} - \frac{29}{16}$

Q

Let $M = \begin{bmatrix} \sin^4 \theta & -1 - \sin^2 \theta \\ 1 + \cos^2 \theta & \cos^4 \theta \end{bmatrix} = \alpha I + \beta M^{-1}$, where $\alpha = \alpha(\theta)$ & $\beta = \beta(\theta)$ are real numbers, and I is the 2×2 identity matrix. If α^* is the minimum of the set $\{\alpha(\theta) : \theta \in [0, 2\pi)\}$ and β^* is the minimum of the set $\{\beta(\theta) : \theta \in [0, 2\pi)\}$, then the value of $\alpha^* + \beta^*$ is

$$\frac{1}{4} \frac{(2 \sin \cos)^2}{\sin^2 \cos^2}$$

$$\sin^2 \theta \cos^2 \theta = t, \quad t \in \left[0, \frac{1}{4}\right]$$

$$\beta = -t^2 - t - 2, \quad 0 \leq t \leq \frac{1}{4}$$

$$\beta(0) = -2$$

$$\beta\left(\frac{1}{4}\right) = -\frac{37}{16} = \beta^*$$

(A) $-\frac{37}{16}$

(B) $-\frac{31}{16}$

(C) $-\frac{17}{16}$

(D) $-\frac{29}{16}$



Let $M = \begin{bmatrix} \sin^4\theta & -1 - \sin^2\theta \\ 1 + \cos^2\theta & \cos^4\theta \end{bmatrix} = \alpha I + \beta M^{-1}$, where $\alpha = \alpha(\theta)$ & $\beta = \beta(\theta)$ are real numbers, and I is the 2×2 identity matrix. If α^* is the minimum of the set $\{\alpha(\theta) : \theta \in [0, 2\pi)\}$ and β^* is the minimum of the set $\{\beta(\theta) : \theta \in [0, 2)\}$, then the value of $\alpha^* + \beta^*$ is

Solution:

$$M = \alpha I + \beta M^{-1}$$

Multiplying both sides by M

$$M^2 = \alpha M + \beta I$$

$$\Rightarrow M^2 - \alpha M - \beta I = 0 \rightarrow \text{characteristic equation} \dots (i)$$

[For a 2×2 matrix $A_{2 \times 2}$, characteristic equation is given by

$$\lambda^2 - [\text{Tr}(A)]\lambda + |A| = 0$$

Solution:

Using (i), we have

$$\alpha = \text{Tr}(M) = \sin^4 \theta + \cos^4 \theta$$

$$\Rightarrow \alpha^* = \frac{1}{2}, \text{ since } \theta \in [0, 2\pi)$$

$$\text{and } -\beta = |M| = \sin^4 \theta \cdot \cos^4 \theta + 2 + \sin^2 \theta \cdot \cos^2 \theta$$

$$\text{Now, } \beta = -(t^2 + t + 2) \quad ; \quad t = \sin^2 \theta \cdot \cos^2 \theta$$

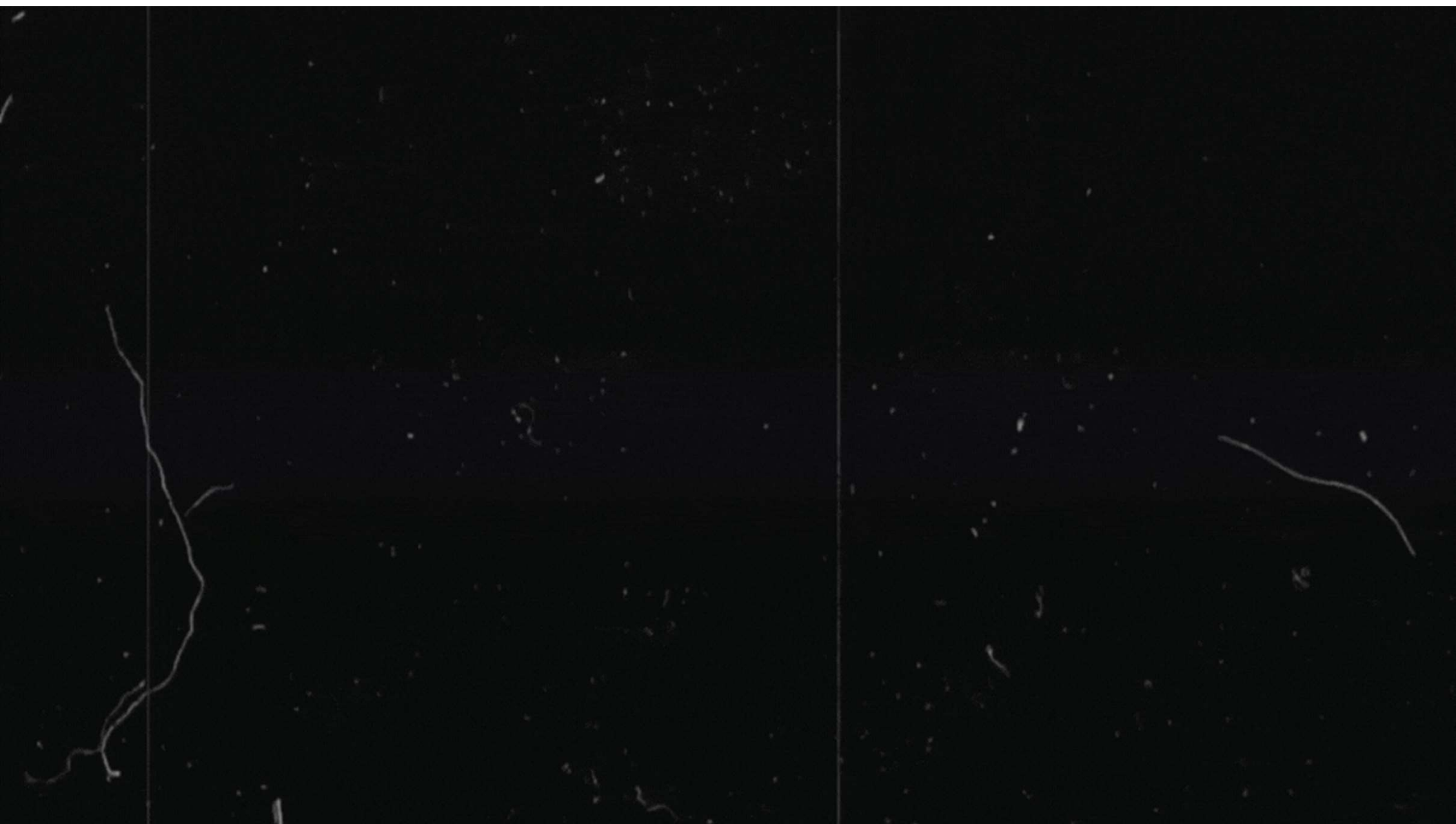
$$0 \leq t \leq \frac{1}{4}$$

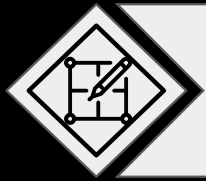
$$\beta(0) = -2$$

$$\beta\left(\frac{1}{4}\right) = -\left(\frac{1}{16} + \frac{1}{4} + 2\right) = \frac{-37}{16}$$

$$\Rightarrow \beta^* = -\frac{37}{16}$$

$$\alpha^* + \beta^* = -\frac{29}{16}.$$





Describing Matrices

Matrix (Expectation)

